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ORIGINAL PAPER

Endogenous enforcement cycles: the co-evolution of tax evasion and audit strategies

Silvia Fedeli[1] · **Elena Costarelli**[1]

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Abstract

This paper studies tax enforcement as a governance problem in which taxpayer compliance and administrative monitoring are jointly endogenous. We formalize this interaction in a two-population evolutionary game between taxpayers, who choose whether to comply or under-declare income, and authority-side enforcement units, which choose whether to inspect. Inspection is costly and imperfectly effective, and the authority side may internalize recoverable but unrecovered tax losses as an institutional-performance loss. The model identifies an interior enforcement benchmark linking tax rates, penalties, audit effectiveness, inspection costs, and institutional accountability. Under canonical replicator dynamics, this benchmark is not a stable endpoint but the center of recurrent compliance–enforcement cycles: evasion raises audit incentives, audits increase compliance, compliance weakens monitoring incentives, and weaker monitoring reopens space for evasion. Numerical illustrations show how enforcement design affects compliance, inspection, recoverable tax losses, net revenue, and institutional performance. The paper highlights that tax-governance outcomes depend not only on sanctions but also on the incentives that sustain monitoring effort over time.

Keywords Tax compliance · Tax evasion · Endogenous enforcement · Audit incentives · Evolutionary games · Replicator dynamics · Enforcement cycles · Fiscal governance

JEL Classification H26 · C73 · K42 · D73 · C61 · H21

✉ Silvia Fedeli
silvia.fedeli@uniroma1.it

Elena Costarelli
ele.costarelli@hotmail.it

[1] Department of Economics and Law, Sapienza University of Rome, Via del Castro Laurenziano 9, 00161 Rome, Italy

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1 Introduction

Enforcement is a governance problem. In many regulatory settings, compliance depends not only on statutory parameters, such as tax rates, penalties, or formal audit rules, but also on whether monitoring is costly to supply and whether the enforcing authority has incentives to sustain it. Tax enforcement is a clear case. A tax administration may deter evasion by inspecting, recovering unpaid taxes, and imposing penalties. Yet, inspection is costly, and its return depends on the behavior it helps to change. If enforcement succeeds in raising compliance, the immediate fiscal return to further inspection may decline. Monitoring may then weaken, and evasion may again become attractive. This paper studies precisely this feedback.

We analyze income tax compliance as the outcome of repeated strategic interaction between two sides of a governance relationship: taxpayers and the authority side of tax administration. Taxpayers choose whether to comply or underdeclare income; authority-side enforcement units choose whether to inspect. The key premise is that both compliance and monitoring are adaptive behaviors. They respond to relative payoffs over time rather than being fixed choices made once and for all under full rationality. This perspective is consistent with a central concern in the economics of governance: enforcement effectiveness depends on endogenous monitoring incentives, not merely on the formal severity of sanctions.

To capture this feedback in a tractable way, we model taxpayers and the authority side as two large populations whose members are randomly and repeatedly matched. Strategies evolve through payoff-based learning: strategies that perform better within each population tend to spread. In this setting, the authority's monitoring decision is explicitly incentive-driven. Inspection may recover unpaid taxes and fines, but it also requires resources. Taxpayer compliance, in turn, depends on the expected payoff from honesty relative to under-declaration, given the likelihood and effectiveness of inspection. The same enforcement instrument, therefore, affects both sides of the interaction: it changes taxpayers' incentives to evade and the authority's incentive to monitor.

The resulting mechanism is as follows. When evasion is widespread, inspection becomes more valuable, because unpaid tax can potentially be recovered and penalties can be collected. Greater inspection reduces the payoff from underdeclaration and therefore supports compliance. Higher compliance, however, also reduces the immediate return to inspection, because there are fewer evaders to detect. The model builds this two-sided incentive feedback into the payoff structure. Whether the resulting adjustment moves toward a boundary regime or generates recurrent movement around an interior benchmark is determined by the evolutionary dynamics analyzed below.

In particular, we derive threshold conditions linking the tax rate, the penalty, audit effectiveness, inspection costs, the intensity of under-declaration, and the authority's institutional concern for recoverable but unrecovered tax losses. These conditions identify when an interior enforcement-compliance benchmark exists and when the system moves toward boundary regimes. The interior benchmark is

economically meaningful, because it summarizes the long-run balance between compliance incentives and monitoring incentives. Under the canonical replicator, however, it is not generally an asymptotically stable destination. Rather, it is the center around which compliance and inspection recurrently move.

The analysis yields several insights that speak directly to governance concerns. First, tax rates matter, because they change both the gain from underdeclaration and the fiscal return to enforcement. Second, penalties and audit effectiveness can improve the compliance benchmark, but their effects on monitoring are not mechanical: once evasion is reduced, the marginal return to inspection may also fall. Third, costly monitoring creates a governance trade-off that is absent when the audit probability is treated as fixed. Fourth, institutional accountability matters. If the authority is evaluated only on immediate cash revenue, noninspection may appear attractive when inspection is costly. If instead the authority internalizes recoverable tax losses as part of its institutional performance, the benchmark level of compliance rises, because noninspection becomes less attractive in the presence of evasion.

These mechanisms are illustrated through numerical simulations. The simulations show how compliance, inspection, recoverable tax losses, net revenue, and institutional performance vary across enforcement scenarios. They also clarify why a static interpretation of the benchmark would be misleading. The relevant policy question is not only where the stationary benchmark lies, but how the system moves around it. A reform that improves the compliance benchmark may still generate periods of weak monitoring if the authority's incentives to inspect decline after compliance improves. Conversely, reforms that strengthen the institutional value of sustained enforcement can reduce the space for recurrent evasion by making monitoring incentives less dependent on immediate recoveries alone.

Overall, this paper contributes to the economics of governance by providing a dynamic account of tax enforcement grounded in endogenous monitoring incentives. Standard deterrence models emphasize tax rates, penalties, and detection probabilities. This paper asks how detection effort itself is sustained when it is costly and when its return depends on taxpayer behavior. Relative to evolutionary optimal-control models of audit policy, the paper does not ask which audit probability a forward-looking regulator should choose; it asks when inspection itself remains a sustainable authority-side strategy as monitoring and compliance co-evolve. The result is a simple but powerful governance mechanism: enforcement policy affects compliance not only by changing the expected cost of evasion, but also by changing the authority's incentive to keep monitoring. The next section situates this perspective within the tax-evasion literature, highlighting how deterrence benchmarks, tax-administration models, and evolutionary approaches motivate a framework in which compliance and monitoring co-evolve through feedback effects.

2 Literature review

The canonical deterrence model of Allingham and Sandmo (1972) treats tax evasion as an individual choice under uncertainty, with the taxpayer choosing how much income to declare given the audit probability, the tax rate, and the penalty

structure. A central implication of their expected-utility framework is that the relation between the tax rate and evasion is not positive. In that model, under the relevant risk-aversion assumptions, a higher tax burden reduces evasion, giving rise to the paradoxical implication that governments may increase compliance by increasing the tax burden. Yitzhaki (1974) clarifies the logic of this result by showing that, when the penalty is levied on the tax evaded rather than on undeclared income, the tax rate affects the risky wealth prospect faced by the taxpayer; under decreasing absolute risk aversion, this channel leads a higher tax rate to reduce evasion. Later qualifications show that this result depends on risk preferences and on the structure of sanctions, and need not hold under risk neutrality or under alternative penalty specifications (see the subsequent survey on this issue by Sandmo 2005, 2012; Alm 2019; Slemrod 2019; Dell'Anno 2022). The lesson for the present paper is that the tax-rate effect should not be treated as a simple one-way comparative static. We study a different governance margin: tax rates, penalties, and audit effectiveness affect compliance not only through the taxpayer's private declaration decision, but also through the authority's incentive to supply costly monitoring.

A second stream of literature brings the analysis closer to this governance margin by treating enforcement as a costly activity carried out by a tax administration under resource constraints. Public enforcement models emphasize that enforcement outcomes depend on the incentives of enforcers and on the organization of monitoring (Becker and Stigler 1974; Polinsky and Shavell 2000). In tax administration, Sandmo (1981), Slemrod and Yitzhaki (1985), Slemrod (1990), and Slemrod and Yitzhaki (2002) stress that enforcement resources are policy instruments and that the optimal size and allocation of a collection agency depend on both costs and taxpayer responses. This perspective motivates the present model's focus on the authority's incentive to supply costly inspection.

Strategic models of tax compliance also show that enforcement and evasion should be analyzed jointly. Greenberg (1984) studies tax compliance as a dynamic game. Reinganum and Wilde (1985) compare audit cut-off policies with random audits. Graetz et al. (1986) incorporate enforcement activity into the taxpayer's decision problem. Erard and Feinstein (1994) introduce taxpayer heterogeneity, while Cronshaw and Alm (1995) consider two-sided uncertainty and show that higher penalties may reduce both cheating and the incentive to inspect. These contributions clarify the strategic nature of enforcement. The present paper complements them by using an evolutionary framework in which population shares adjust through payoff-based selection rather than by solving a full-rationality equilibrium with fixed audit rules.

Empirical and experimental evidence reinforces the importance of behavioral responses, social interaction, and perceptions of enforcement. Laboratory and field studies show that audits, penalties, tax rates, moral suasion, intrinsic motivation, and peer effects can all matter for compliance (Baldry 1987; Alm et al. 1990, 1992a, 1992b, 1993, 2017; Pommerehne and Weck-Hannemann 1996; Ali et al. 2001; Torgler 2003; Dwenger et al. 2016). Meta-evidence from tax-evasion games also suggests that behavioral responses often diverge from simple deterrence predictions (Malezieux 2018; Alm and Malezieux 2021). These findings motivate models in

which behavior adapts over time and in which enforcement cannot be treated as a mechanical fixed input.

Agent-based and evolutionary approaches move in this direction. Agent-based models of tax compliance represent heterogeneous taxpayers, social interaction, and administrative rules explicitly (Bloomquist 2006; Hashimzade et al. 2015). Evolutionary game models have also been used to study fraud and compliance in consumption-tax settings (Chica et al. 2021). In evolutionary tax-evasion models, De Giovanni et al. (2019) combine prospect-theory taxpayers with an optimal-control problem in which a forward-looking tax authority chooses audit probabilities to maximize intertemporal net tax revenues. De Giovanni et al. (2020) extend this framework to heterogeneous modes of evasion, distinguishing mild and aggressive evaders and studying the allocation of audit probabilities under resource constraints. These contributions are close to the present paper, because they make auditing endogenous, but they do so through optimal regulation rather than through an evolutionary authority-side strategy.

De Giovanni et al. (2023) study a different but also related two-population setting in which taxpayers interact with auditors who may be honest or corrupt and may request bribes. Their central mechanism concerns the joint evolution of tax evasion, auditor corruption, social norms, and corruption cycles. The present paper asks a different governance question. It abstracts from bribery, auditor corruption, prospect-theory probability weighting, and an optimizing regulator. Inspection and non-inspection are instead modeled as authority-side strategies whose population shares evolve according to relative payoffs. This allows the analysis to separate actual fiscal revenue from the authority's institutional-performance payoff, derive closed-form enforcement thresholds, and interpret the stationary benchmark as the center and time-average of recurrent compliance–enforcement motion under the canonical replicator.

Evolutionary game theory provides the formal language for this type of adaptation. Replicator dynamics describe how strategies that yield higher payoffs spread within a population and have been widely used to model imitation, learning, and payoff-based selection (Schuster and Sigmund 1983; Weibull 1995; Samuelson and Zhang 1992; Friedman 1998; Schlag 1998; Mailath 1998; Hofbauer and Sigmund 2003; Sandholm 2010; Cressman and Tao 2014). In the present setting, this approach is useful, because both sides of the enforcement relationship adjust over time: taxpayers respond to the relative payoff of compliance and evasion, while authority-side units respond to the relative payoff of inspection and non-inspection. The dynamics, therefore, capture the feedback between compliance and monitoring that is central to the governance problem studied here.

Finally, the paper speaks to the economics of governance by emphasizing how public organizations evaluate and sustain enforcement effort. Tax administrations do not operate only as mechanical revenue collectors. They face costs, capacity constraints, political and administrative objectives, and performance criteria that may include cash collections, recovered liabilities, deterrence, and the reduction of recoverable tax losses. This perspective is consistent with governance work on bureaucratic incentives and the limits of purely mechanical performance criteria (Prendergast 2003). It motivates a distinction between actual fiscal revenue and

the institutional payoff associated with enforcement performance. The model below introduces this distinction in reduced form and studies how it affects the co-evolution of compliance and monitoring.

3 Model

We model tax compliance as a two-population evolutionary game between taxpayers and the authority side of tax administration. Taxpayers may either comply fully with the tax law or under-declare income. The authority side is represented by enforcement units or administrative routines that may either inspect or not inspect. The model is intentionally parsimonious: it abstracts from risk-based targeting, heterogeneous income, and explicit belief updating in order to isolate the feedback between taxpayer compliance and the authority's incentive to supply costly monitoring.

Let Y denote true income, which is initially assumed to be common across taxpayers and not directly observed by the authority. The statutory tax rate is τ, with $0 < \tau < 1$. A cheating taxpayer declares only a fraction β of true income, with $0 \leq \beta < 1$; an honest taxpayer declares the full income. If evasion is detected, the authority can recover a fraction α of the unpaid tax and associated penalty base, where $0 < \alpha \leq 1$. In the current specification, α is interpreted as the effective detection and recovery rate conditional on inspection. It is not a belief variable and does not evolve endogenously in the baseline model. The penalty rate on evaded income is $\varphi > 0$. Inspection costs are proportional to the tax liability on true income and equal to $C\,\tau\,Y$, with $C > 0$.

The taxpayer population contains a share q_H of honest taxpayers and a share $q_C = 1 - q_H$ of cheating taxpayers. The authority side is represented by a population of enforcement units or administrative routines: a share p_I inspect and a share $p_{NI} = 1 - p_I$ do not inspect.

3.1 Payoffs

The payoff matrix is built from the four possible matches between the two populations. A taxpayer may be honest or cheating, and the authority may inspect or not inspect. Each random match therefore belongs to one of four cases: an inspecting unit matched with an honest taxpayer; an inspecting unit matched with a cheating taxpayer; a non-inspecting unit matched with an honest taxpayer; and a non-inspecting unit matched with a cheating taxpayer. These cases define the payoffs from which the fitness functions and the replicator dynamics are derived.

First, consider an inspecting unit matched with an honest taxpayer. The taxpayer has declared the entire income, so there is no evasion to detect. The authority receives the tax due on true income but bears the inspection cost

$$\pi_A(I, H) = \tau Y - C\tau Y. \tag{1}$$

The honest taxpayer pays the tax due and obtains disposable income

$$\pi_T(I,H) = (1-\tau)Y. \tag{2}$$

In this case, audit effectiveness does not affect payoffs, because the taxpayer has already declared the full tax base.

Second, consider an inspecting unit matched with a cheating taxpayer. The taxpayer has declared only a fraction β of true income. If an inspection is carried out, the authority does not necessarily recover the entire evaded amount. The parameter α captures the effectiveness of the inspection: only the fraction α of evaded tax and related fines is detected and recovered. The authority's payoff is therefore the tax paid on declared income, plus the effectively recovered tax and penalty, net of inspection cost

$$\pi_A(I,C) = \beta\tau Y + \alpha\tau(1-\beta)Y + \alpha\varphi(1-\beta)Y - C\tau Y. \tag{3}$$

The cheating taxpayer's payoff under inspection is

$$\pi_T(I,C) = Y - \beta\tau Y - \alpha\tau(1-\beta)Y - \alpha\varphi(1-\beta)Y. \tag{4}$$

Third, consider a non-inspecting unit matched with an honest taxpayer. Since the taxpayer has declared the entire income, the authority receives the tax due without incurring any inspection cost

$$\pi_A(NI,H) = \tau Y, \tag{5}$$

and the taxpayer obtains the same after-tax income as in the inspected honest case

$$\pi_T(NI,H) = (1-\tau)Y. \tag{6}$$

Fourth, consider a non-inspecting unit matched with a cheating taxpayer. This is the only case in which evasion is neither detected nor recovered. The authority collects only the tax paid on the declared fraction of income, τ β Y. This is actual fiscal revenue. However, non-inspection may also be costly from an institutional-governance perspective if the administration is evaluated not only on current collections but also on its ability to limit recoverable tax losses. We capture this by allowing the authority's evolutionary payoff to include an institutional-performance loss from recoverable but unrecovered tax losses

$$\pi_A(NI,C) = \tau\beta Y - \lambda\alpha\tau(1-\beta)Y. \tag{7}$$

Equation (7) assigns the non-inspecting unit a payoff lower than the cash revenue τβY collected from a cheating taxpayer. The difference is not an additional monetary outflow, but a reduced-form institutional penalty attached to recoverable but unrecovered tax losses. The parameter λ can therefore be interpreted as a reduced-form weight attached to these losses in the authority-side performance payoff; it may reflect accountability mechanisms, such as recovery targets, reporting on recoverable losses, parliamentary scrutiny, or internal performance evaluation, without treating the losses as direct accounting costs. Thus, τβY is the cash revenue actually collected, while λατ(1−β)Y measures the institutional value of the tax loss that an effective inspection would have recovered. When λ=1, the authority fully

internalizes these recoverable but unrecovered tax losses in its payoff. When λ=0, the authority's payoff coincides with cash revenue (this case is analyzed in appendix B). The qualitative dynamic result of the paper does not depend on this institutional-performance term. Appendix B derives the pure cash-revenue case in full, where the term is removed from the non-inspection payoff, and shows that the recurrent two-population structure is unchanged.

The cheating taxpayer who is not inspected pays tax only on declared income and keeps the remaining income

$$\pi_T(NI, C) = Y - \beta\tau Y. \tag{8}$$

The non-inspection cases also capture the fact that, in real tax systems, there is generally no one-to-one correspondence between taxpayers and inspectors. The number of taxpayers is typically much larger than the enforcement capacity of the authority. Non-inspection therefore represents not only a deliberate strategic choice in a single match, but also the ordinary consequence of limited administrative capacity.

The four possible matches define the payoff structure of the game. In each match, there is one payoff for the taxpayer and one payoff for the authority, so the model has eight payoff entries in total. Since taxpayers and authority-side units are randomly matched, the payoff from each strategy is an expected payoff. For example, the payoff from inspection depends on whether the inspecting unit meets an honest taxpayer or a cheating taxpayer, and therefore on the share of honest and cheating taxpayers in the population. Similarly, the payoff from honesty or cheating depends on the probability of being matched with an inspecting or non-inspecting unit. These expected strategy payoffs are the fitness functions that drive the replicator dynamics.

3.2 Fitness functions

Expected payoffs, or fitnesses, are obtained by weighting each payoff by the composition of the other population. For compactness, write: $x = q_H, 1 - x = q_C, y = p_I, 1 - y = p_{NI}$.

The authority's expected payoff from inspection is

$$f_A(I) = x\pi_A(I, H) + (1 - x)\pi_A(I, C).$$

The authority's expected payoff from non-inspection is

$$f_A(NI) = x\pi_A(NI, H) + (1 - x)\pi_A(NI, C).$$

The taxpayer's expected payoff from honesty is

$$f_T(H) = y\pi_T(I, H) + (1 - y)\pi_T(NI, H).$$

The taxpayer's expected payoff from cheating is

$$f_T(C) = y\pi_T(I, C) + (1 - y)\pi_T(NI, C).$$

The two payoff differences that drive the dynamics are

$$\Delta_T(y) = f_T(H) - f_T(C) = (1-\beta)Y[\alpha(\tau+\varphi)y - \tau] \tag{9}$$

and

$$\Delta_A(x) = f_A(I) - f_A(NI) = Y\{\alpha(1-\beta)[\varphi + (1+\lambda)\tau](1-x) - C\tau\}. \tag{10}$$

Equation (9) shows that taxpayer behavior depends on effective inspection pressure, α y, and on the combined tax and penalty loss from detection, τ+φ. Equation (10) shows that the authority's monitoring incentive increases with the share of cheaters, audit effectiveness, penalties, and the institutional internalization of the recoverable tax losses. The parameter λ affects the authority's incentive to inspect; it does not directly affect the taxpayer indifference condition.

3.3 Replicator dynamics

The two-population replicator is

$$\dot{x} = x(1-x)\Delta_T(y) \tag{11}$$

$$\dot{y} = y(1-y)\Delta_A(x). \tag{12}$$

Substituting (9) and (10), the dynamic system is

$$\dot{x} = x(1-x)(1-\beta)Y\left[\alpha(\tau+\varphi)y - \tau\right] \tag{13}$$

$$\dot{y} = y(1-y)Y\{\alpha(1-\beta)[\varphi + (1+\lambda)\tau](1-x) - C\tau\}. \tag{14}$$

Equations (13) and (14) are the canonical continuous-time replicator equations for the two-population game. The logarithmic rate form is equivalent only along interior trajectories, where all shares are strictly positive. Because boundary behavior is central to the analysis, we use the canonical form throughout.

4 Analytical results

Proposition 1. Interior stationary benchmark
An interior stationary point exists if and only if

$$\alpha(\tau+\varphi) > \tau \tag{15}$$

and

$$\alpha(1-\beta)[\varphi + (1+\lambda)\tau] > \tau\,\mathrm{C}. \tag{16}$$

When these inequalities hold, the interior stationary point is

$$p_I^* = \tau/[\alpha(\tau+\varphi)] \tag{17}$$

$$p_{NI}^* = 1 - p_I^* = 1 - {}^{\tau}/_{[\alpha(\tau+\varphi)]}, \tag{18}$$

$$q_C^* = \tau C/\{\alpha(1-\beta)[\varphi + (1+\lambda)\tau]\} \tag{19}$$

$$q_H^* = 1 - q_C^* = 1 - \tau C/\{\alpha(1-\beta)[\varphi + (1+\lambda)\tau]\}. \tag{20}$$

Proof. An interior stationary point requires $\Delta_T(y) = 0$ and $\Delta_A(x) = 0$. Solving (9) for y gives p_I^*. Since $p_{NI} = 1 - p_I$, this immediately gives p_{NI}^*. Solving (10) for 1−x gives q_C^*, and therefore q_H^*. Conditions (15)–(16) ensure that p_I^*, p_{NI}^*, q_C^*, and q_H^* all lie strictly between zero and one.

Proposition 1 defines the interior benchmark of the model. This benchmark has a straightforward economic meaning. On the taxpayer side, the share of inspecting units is such that under-declaration no longer gives a payoff advantage over full compliance. On the authority side, the share of cheating taxpayers is such that the expected return from inspection is exactly balanced by inspection costs and by the payoff from non-inspection. Thus, the benchmark is the point at which enforcement pressure is sufficient to discipline taxpayers, while the amount of evasion is sufficient to justify inspection. This does not mean that the system necessarily converges to that point; it only identifies the stationary balance between compliance incentives and monitoring incentives. Whether the system converges to this benchmark is a separate dynamic question, addressed below. The model assumes $\beta<1$, because β measures the fraction of income declared by a cheating taxpayer. If $\beta=1$, the taxpayer declares all income and there is no under-declaration. The cheating strategy therefore disappears.

4.1 Comparative statics

Table 1 summarizes the comparative statics of the interior benchmark. The table reports the direction of change in the stationary inspection share p_I^* and in the stationary cheating share q_C^*. The effects on p_{NI}^* and q_H^* are the opposite, because $p_{NI}^* = 1 - p_I^*$ and $q_H^* = 1 - q_C^*$. All signs are ceteris paribus: in each row, the listed parameter changes, while the other parameters are held fixed.

The comparative statics should be read as changes in the stationary benchmark of the two-population game. The benchmark should not be interpreted as a point of automatic convergence. It is the population composition at which the payoff difference between the two taxpayer strategies and the payoff difference between the two authority strategies are both zero.

The value of p_I^* is determined by the taxpayer side of the interaction. It is the inspection share that balances the relative payoff of honesty and under-declaration in the taxpayer population. When p_I is below this level, cheating is relatively more attractive; when inspection is sufficiently widespread, honesty can be sustained. The comparative statics of p_I^* therefore show how much inspection activity must be present in the authority-side population to sustain the stationary taxpayer composition.

Table 1 Comparative statics of the interior benchmark

Param	Effect on p_I^* (inspection)	Effect on q_C^* (cheating)	Economic interpretation
τ	>0	>0	A higher tax rate requires more inspection to deter under-declaration and, because inspection costs rise with tax liability, more cheating to sustain inspection
α	<0	<0	More effective audits strengthen deterrence and increase the return to inspection; both the inspection share and the cheating share required at the benchmark fall
φ	<0	<0	A higher penalty strengthens deterrence and the return from detected evasion; both benchmark shares fall
C	No direct effect	>0	Higher inspection costs require more cheating before inspection becomes worthwhile; taxpayers' incentives are unaffected directly
β	No direct effect	>0	A larger declared fraction reduces the recoverable base per cheater, so more cheating is needed to sustain inspection
λ	No direct effect	<0	Stronger accountability makes non-inspection less attractive, so less cheating is needed to sustain inspection; the taxpayer-side inspection threshold is unchanged

The value of q_C^* is determined by the authority side of the interaction. It is the cheating share that balances inspection and non-inspection in the authority-side population. If inspection becomes more costly, or if each cheating taxpayer leaves less recoverable income, a larger cheating share is needed for inspection to remain attractive. Conversely, when audit effectiveness or penalties increase, inspection becomes more rewarding and the same monitoring incentive can be sustained with a lower share of cheaters.

The parameter λ affects this second margin only. It changes the authority's evaluation of non-inspection when recoverable but unrecovered tax losses are present, but it does not enter the taxpayer payoff comparison directly. A higher λ therefore reduces q_C^* and raises q_H^*, while leaving p_I^* unchanged.

The tax rate affects both populations. It increases the equilibrium inspection proportion required to sustain honesty among taxpayers. At the same time, in the authority condition, it raises q_C^*: although a higher tax rate increases recoverable tax, it also raises the inspection-cost term $C\tau Y$, and the net effect in the equilibrium condition is positive. This result concerns the equilibrium distribution of strategies in the present evolutionary model, not a general tax-rate prediction from fixed-audit deterrence models. Appendix A reports the derivatives behind (Table 1).

Proposition 2. Linearization around the interior benchmark
Suppose that conditions (15) and (16) hold, so that the interior benchmark exists. Under the canonical two-population replicator, this benchmark is a rest point of the system, but it is not an attracting rest point. If the system starts close to the benchmark, the interaction between taxpayer behavior and inspection incentives does not necessarily drive it back monotonically to that point. Instead, deviations activate a feedback mechanism: a higher share of cheating taxpayers makes inspection more attractive; more inspection makes cheating less attractive; higher compliance then weakens the return to inspection; and weaker inspection can again make cheating more attractive.

Formally, the linearized dynamics around the interior benchmark are rotational rather than convergent.

Proof. Let $x = q_H$ denote the share of honest taxpayers and $y = p_I$ the share of inspecting units. Define

$$A = (1 - \beta)\alpha(\tau + \varphi)Y$$

$$B = \alpha(1 - \beta)[\varphi + (1 + \lambda)\tau]Y.$$

Around the interior benchmark (x^*, y^*), the replicator system can be written as

$$\dot{x} = x(1 - x)A(y - y^*)$$

$$\dot{y} = y(1 - y)B(x^* - x).$$

The first equation says that the share of honest taxpayers increases when inspection is above its benchmark level. The second equation says that the share of inspecting units increases when the share of honest taxpayers is below its benchmark level, that is, when cheating is sufficiently widespread.

The Jacobian at (x^*, y^*) is

$$J(x^*, y^*) = \begin{bmatrix} 0 & x^*(1-x^*)A \\ -y^*(1-y^*)B & 0 \end{bmatrix}.$$

Its trace is

$$trJ(x^*, y^*) = 0,$$

and its determinant is

$$\det J(x^*, y^*) = x^*(1-x^*)y^*(1-y^*)AB > 0.$$

The eigenvalues are therefore

$$\mu_{\{1,2\}} = \pm i\sqrt{\left[x^*(1-x^*)y^*(1-y^*)AB\right]}.$$

The trace is zero and the determinant is positive. The eigenvalues are purely imaginary. Hence, the interior benchmark is not locally asymptotically stable under the canonical replicator. Economically, it should be interpreted as the balance point of the compliance–enforcement interaction, not as a stable endpoint to which the system necessarily converges.

Proposition 3. Interior trajectories around the benchmark
Suppose that conditions (15) and (16) hold, so that the interior benchmark exists. If the initial distribution of taxpayers and authority-side units is strictly interior, the canonical replicator does not generally drive the system to the benchmark. Instead, the shares of honest taxpayers and inspecting units move around it.

The reason is that the two populations continuously react to each other. When cheating is high, inspection becomes more attractive. When inspection is high, cheating becomes less attractive and honesty increases. When honesty becomes widespread, inspection becomes less rewarding. When inspection falls, cheating can become attractive again. The result is a recurrent compliance–enforcement movement around the interior benchmark.

This motion can be represented by a quantity that remains constant along each interior trajectory

$$H(x, y) = B\left[x^* \ln x + (1-x^*)\ln(1-x)\right] + A\left[y^* \ln y + (1-y^*)\ln(1-y)\right]. \quad (21)$$

Equation (21) is a first integral of the replicator system: it assigns a constant value to each interior trajectory. Therefore, along the canonical replicator

$$dH/dt = 0. \quad (22)$$

Equation (22) means that, once the system starts from an interior point, it remains on the same trajectory around the benchmark. Different initial conditions generate different interior paths, but the benchmark remains the central reference point of the motion. Thus, the model predicts recurrent adjustment in compliance and inspection, rather than monotone convergence to an asymptotically stable endpoint.

The level-curve interpretation also gives the recurrence a formal basis. The function H is strictly concave on the interior of the unit square and tends to $-\infty$ at the boundary.

Hence, each non-stationary level set around the interior benchmark is compact and closed. Since the interior benchmark is the only rest point on such a level set, the corresponding interior path is a closed periodic orbit under the canonical replicator.

Proof. Differentiating (21) gives

$$\frac{dH}{dt} = B\left[\frac{x^*}{x} - \frac{(1-x^*)}{1-x}\right]\dot{x} + A\left[\frac{y^*}{y} - \frac{(1-y^*)}{1-y}\right]\dot{y}.$$

Substituting the dynamics from Proposition 2 yields

$$\frac{dH}{dt} = AB(x^* - x)(y - y^*) + AB(y^* - y)(x^* - x) = 0.$$

Thus, H is constant along interior trajectories. Different initial conditions select different level curves, but they do not change the interior benchmark itself.

Proposition 4. The interior stationary benchmark as the average composition of a cycle

Suppose that conditions (15) and (16) hold, so that the interior stationary benchmark exists. Let $x = q_H$ denote the share of honest taxpayers and $y = p_I$ the share of inspecting units. The interior stationary benchmark is the pair (x^*, y^*) at which the two population shares would stop changing if the system were exactly at that point.

Now, consider a complete interior cycle. By this, we mean a path in which the model remains inside the economically relevant case where both taxpayer strategies and both authority-side strategies are present. Along the cycle, there are always some honest taxpayers, some cheating taxpayers, some inspecting units, and some non-inspecting units. After a finite time T, the system returns to the same composition of taxpayers and authority-side units from which it started. Along such a cycle, the average share of honest taxpayers and the average share of inspecting units are equal to the stationary benchmark values

$$\bar{x} = \left(\frac{1}{T}\right)\int_0^T x(t)dt = x^*$$

$$\bar{y} = \left(\frac{1}{T}\right)\int_0^T y(t)dt = y^*.$$

This result gives the stationary benchmark a precise economic interpretation. The benchmark is not necessarily the point at which the system settles. Rather, it is the average composition around which the cycle moves. Periods in which honesty is above its benchmark value are offset by periods in which honesty is below it. Similarly, periods of stronger inspection are offset by periods of weaker inspection. Over a full cycle, these deviations cancel out, so the time averages coincide with x^* and y^*

Proof. Around the interior stationary benchmark, the replicator dynamics imply

$$\frac{d}{dt}\ln\left[\frac{x}{1-x}\right] = A(y - y^*)$$

$$\frac{d}{dt}\ln\left[\frac{y}{1-y}\right] = B(x^* - x).$$

Over a full cycle of length T, the system returns to its initial values: x(T)=x(0) and y(T)=y(0). Therefore, the log-odds terms ln[x/(1-x)] and ln[y/(1-y)] also return to their initial values. Integrating the two equations over the full cycle gives

$$\int_0^T \left[y(t) - y^*\right]dt = 0$$

$$\int_0^T \left[x^* - x(t)\right]dt = 0.$$

Dividing by T gives

$$\bar{y} = y^*$$

$$\bar{x} = x^*.$$

Thus, even when the system does not converge to the stationary benchmark, the benchmark remains economically meaningful: it is the average distribution of compliance and inspection over the interior cycle.

The same argument also justifies using the benchmark to summarize the average revenue components along a closed interior cycle. Audit revenue and recoverable but unrecovered tax losses depend on products of taxpayer and authority-side shares. Over a full cycle, the following product averages hold:

$$\left(\frac{1}{T}\right)\int_0^T (1 - x(t))y(t)dt = (1 - x^*)y^*$$

$$\left(\frac{1}{T}\right)\int_0^T (1 - x(t))(1 - y(t))dt = (1 - x^*)(1 - y^*).$$

The first identity gives the average share of cheating taxpayers who are inspected. The second gives the average share of cheating taxpayers who are not inspected. Hence, the stationary benchmark can be used as a compact revenue-accounting benchmark for closed interior cycles, not as a claim that the system converges to that point.

4.2 Interpretation of the cycle

The baseline replicator does not imply monotone convergence to the stationary benchmark. When the interior conditions are satisfied, the benchmark is instead the point around which the interaction between compliance and inspection is organized. If the share of cheating taxpayers is high, inspection becomes more profitable, because more unpaid tax and penalties can potentially be recovered. As inspection expands, underdeclaration loses payoff value and compliance is supported. As compliance becomes more widespread, however, the immediate return to inspection declines. If inspection then weakens, underdeclaration may again become attractive. This is the economic content of the center result: the model captures an endogenous feedback between taxpayer behavior and monitoring incentives.

This center property follows from the canonical two-population replicator used in the baseline model. It should not be interpreted as a claim that all possible adjustment processes generate exactly the same motion. Additional frictions, such as mutation, noise, discrete-time adjustment, or policy rules that impose minimum inspection levels, may change the stability properties of the dynamics. The main economic message is therefore not the exact geometry of the trajectories, but the feedback mechanism: compliance and inspection incentives are jointly determined, and their long-run behavior depends on both the payoff structure and the adjustment process.

Proposition 5. Boundary regimes
The interior stationary benchmark exists only if all four strategies remain present in the long-run composition of the two populations. This means that, in the taxpayer population, there must be both honest taxpayers and cheating taxpayers; and, in the authority-side population, there must be both inspecting and non-inspecting units. Formally, this requires all benchmark shares to lie strictly between zero and one.

Conditions (15) and (16) are exactly the conditions that make this possible. If they hold, the model has an interior benchmark: q_H^*, q_C^*, p_I^*, and p_{NI}^* are all positive and smaller than one. If one of these conditions fails, at least one of these shares would fall outside the admissible interval. In that case, the formulas no longer describe an economically meaningful interior composition. The relevant outcome is instead a boundary regime, where one strategy disappears from one of the two populations.

This point is important, because the replicator dynamics do not create a strategy once its share has become zero. If all taxpayers are cheating, the share of honest taxpayers cannot increase under the canonical replicator unless honesty is reintroduced from outside the model. Similarly, if no authority inspects, inspection cannot reappear endogenously once its share is exactly zero. This is why the boundaries are absorbing under the canonical replicator.

The two margins behind the interior benchmark are

$$\mathrm{D} = \alpha(\tau + \varphi) - \tau$$

and

$$\mathrm{E} = \alpha(1 - \beta)[\varphi + (1 + \lambda)\tau] - \mathrm{C}\tau.$$

The first margin, D, refers to the taxpayer side. It compares the effective expected cost of cheating under inspection with the tax saving from under-declaration. If D>0, inspection can make cheating unattractive enough for honesty to survive in the taxpayer population. If D<0, cheating remains too attractive even under inspection. The model then moves toward a low-compliance boundary, because the honest strategy cannot be sustained.

The second margin, E, refers to the authority side. It compares the payoff gain from inspecting a cheating taxpayer with the cost of inspection. If E>0, inspection can be worthwhile when cheating is present. If E<0, inspection incentives are too weak: even when evasion exists, the authority does not gain enough from inspection to cover its cost. The model then moves toward a low-inspection boundary, because the inspection strategy cannot be sustained.

Thus, when D>0 and E>0, both populations can contain both strategies, and the interior benchmark exists. When D<0, the taxpayer population tends toward a boundary with very low compliance. When E<0, the authority-side population tends toward a boundary with very low inspection. Equality cases, D=0 or E=0, are knife-edge cases: they separate regimes but do not describe robust economic outcomes.

The formulas for p_I^*, p_{NI}^*, q_C^*, and q_H^* should therefore be used as interior benchmark values only when conditions (15) and (16) hold. If these conditions fail, the formulas are useful diagnostically: they tell us why the interior benchmark is not feasible and which side of the model pushes the system toward a boundary regime (Table 2).

5 Numerical illustration

The numerical illustration uses the canonical replicator in (13) and (14). Its purpose is illustrative. The analytical results above establish the cyclic structure and the time-average interpretation of the benchmark; the numerical section shows how the mechanism operates under a simple calibration. A simple relaxation rule of the form

$$\dot{x} = k(x^* - x)$$

$$\dot{y} = k(y^* - y)$$

converges by construction and is not a numerical proxy for the canonical replicator. Accordingly, the numerical illustration reports direct replicator simulations from strictly interior initial conditions.

The baseline numerical parameters are: $Y = 1, \tau = 0.30, \beta = 0.75, C = 0.50$.

We consider four combinations of audit effectiveness and penalties

Scenario 1: α=0.40, φ=0.50

Scenario 2: α=0.90, φ=1.50

Scenario 3: α=0.90, φ=0.50

Scenario 4: α=0.40, φ=1.50.

For each scenario, we consider λ=0, λ=0.5, and λ=1. The case λ=0 is the pure cash-revenue benchmark. The case λ=1 is the full-internalization case.

Table 2 Boundary regimes

Condition	Dynamic implication from interior states	Economic interpretation
$D>0, E>0$	Interior recurrent motion around the benchmark	Mixed compliance–enforcement regime
$D>0, E<0$	Inspection becomes unsustainable; the system moves toward $y\to 0$ and then $x\to 0$	Low-compliance/no-inspection boundary
$D<0, E>0$	Honesty is not viable; the system moves toward $x\to 0$ and then $y\to 1$	Low-compliance/high-inspection boundary
$D<0, E<0$	Both honesty and inspection are unsustainable; the system moves toward $x\to 0$ and $y\to 0$	Collapse of compliance and inspection
$D=0$ or $E=0$	Knife-edge transition	Non-robust boundary between regimes

5.1 Stationary benchmarks

Table 3 reports the candidate stationary benchmark implied by (17)–(20), together with an indicator of whether the candidate is economically admissible as an interior benchmark. The entries should be read as population shares: q_H^* is the share of honest taxpayers, q_C^* is the share of cheating taxpayers, and p_I^* is the share of inspecting units.

The table has two main messages. First, audit effectiveness and penalties are not secondary calibration details: they determine whether an interior compliance–enforcement regime can be sustained. Scenario 2, where both α and φ are high, has the strongest compliance benchmark and the lowest inspection share required by the taxpayer condition. Scenario 1, where both are low, never becomes an interior case, even when λ=1. Second, α and φ enter both sides of the model, but not in exactly the same way. The taxpayer margin depends on α(τ+φ), while the authority margin depends on α(1-β)[φ+(1+λ)τ] relative to Cτ. This explains why Scenarios 3 and 4 have the same p_I^*: in both cases α(τ+φ) is the same. At the same time, the response of q_H^* to λ is stronger in Scenario 3 than in Scenario 4, because λ is multiplied by α in the authority's inspection payoff.

The role of λ is therefore important but limited. A higher λ makes non-inspection less attractive when recoverable but unrecovered tax losses are present, so it raises q_H^* in the interior scenarios. However, λ does not change p_I^*, and it cannot compensate for a weak enforcement technology combined with a low penalty. In Scenario 1 the candidate q_H^* remains negative for all three values of λ, so the formulas diagnose a boundary regime rather than an interior equilibrium.

5.2 Replicator trajectories

The numerical trajectories are obtained by direct integration of the canonical replicator system in (13) and (14). They show how the population shares of honest

Table 3 Candidate stationary benchmarks and interior admissibility

λ	Scenario	α	φ	q_H^*	q_C^*	p_I^*	Interior
0.0	Scenario 1	0.40	0.50	− 0.875	1.875	0.937	no
0.0	Scenario 2	0.90	1.50	0.630	0.370	0.185	yes
0.0	Scenario 3	0.90	0.50	0.167	0.833	0.417	yes
0.0	Scenario 4	0.40	1.50	0.167	0.833	0.417	yes
0.5	Scenario 1	0.40	0.50	− 0.579	1.579	0.937	no
0.5	Scenario 2	0.90	1.50	0.658	0.342	0.185	yes
0.5	Scenario 3	0.90	0.50	0.298	0.702	0.417	yes
0.5	Scenario 4	0.40	1.50	0.231	0.769	0.417	yes
1.0	Scenario 1	0.40	0.50	− 0.364	1.364	0.937	no
1.0	Scenario 2	0.90	1.50	0.683	0.317	0.185	yes
1.0	Scenario 3	0.90	0.50	0.394	0.606	0.417	yes
1.0	Scenario 4	0.40	1.50	0.286	0.714	0.417	yes

taxpayers and inspecting units evolve from alternative initial compositions. The exercise is illustrative: the analytical results above establish the structure of the dynamics, while the simulations show how that structure operates under the calibration used in Sect. 5.1.

For Scenarios 2–4, the stationary composition is admissible: both taxpayer strategies and both authority strategies remain present. The system does not move monotonically toward the stationary composition. Instead, it generates recurrent compliance–enforcement movements around it. This behavior follows from the feedback between the two populations. When cheating is sufficiently widespread, inspection becomes more rewarding for the authority. When inspection becomes more common, cheating becomes less attractive for taxpayers and honesty increases. As compliance rises, inspection becomes less profitable. When inspection then weakens, cheating can again become attractive. The result is a cycle of adjustment rather than convergence to a stable long-run composition.

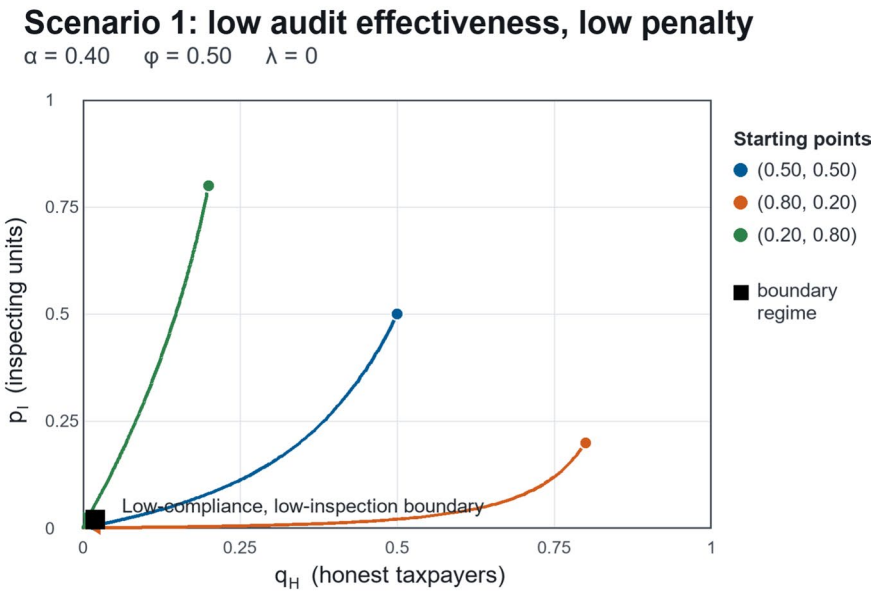
The calibration affects where this recurrent movement is centered. Scenario 2, where both audit effectiveness α and the penalty φ are high, produces the strongest compliance benchmark: the stationary honest share is relatively high and the inspection share required to discipline cheating is relatively low. Scenarios 3 and 4 are weaker interior cases. Scenario 3 has high audit effectiveness but a low penalty; Scenario 4 has a high penalty but lower audit effectiveness. These mixed cases show that penalties and audit effectiveness are not perfect substitutes. A high penalty is less effective when inspections recover little, while effective inspections are less deterrent when the penalty is low.

The accountability parameter λ affects the authority side of the adjustment. A higher λ makes non-inspection less attractive when evasion would have been recoverable, and therefore shifts the stationary composition toward a higher honest share in the interior scenarios. This effect is more pronounced when audit effectiveness is high, because the institutional loss from non-inspection is larger when inspections are capable of recovering more evasion. By contrast, λ does not enter the taxpayer indifference condition and therefore does not change the stationary inspection share p_I^*.

Scenario 1 is qualitatively different. In that calibration, both α and φ are low, so the interior condition fails. The system no longer supports a mixed compliance–enforcement composition with both taxpayer strategies and both authority strategies present. Starting from interior initial conditions, the dynamics move toward a boundary regime characterized by very low compliance and very low inspection. This is not a general prediction about all enforcement systems; it is the implication of a weak-enforcement calibration in which inspection is not productive enough to sustain itself.

Thus, the numerical exercise clarifies the economic mechanism of the model. Productive inspection and sufficiently strong penalties can sustain recurrent compliance–enforcement dynamics around an interior stationary composition. When enforcement is too weak, the same replicator mechanism pushes the system toward a boundary regime. Figure 1 reports two representative trajectories. Panel A shows Scenario 1, where low audit effectiveness and a low penalty make the interior benchmark inadmissible and the dynamics move toward a low-compliance, low-inspection

Panel A. Scenario 1: boundary regime



Panel B. Scenario 2: recurrent motion

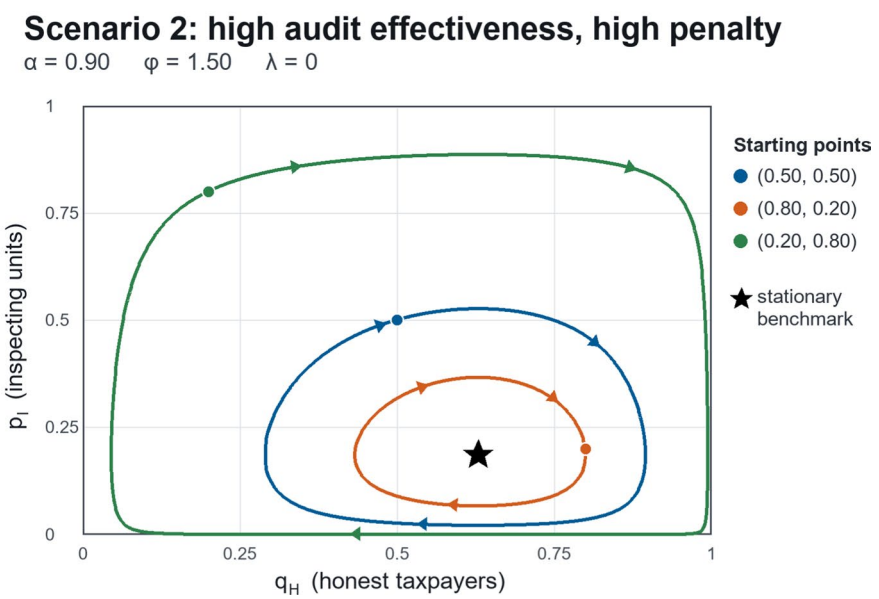


Fig. 1 Representative replicator trajectories. Note that Panel **A** reports Scenario 1 ($\lambda = 0$, $\alpha = 0.40$, $\varphi = 0.50$), where the interior benchmark is not admissible and the dynamics move toward the low-compliance, low-inspection boundary. Panel **B** reports Scenario 2 (λ=0, α=0.90, φ=1.50), where trajectories move around the interior benchmark

boundary. Panel B shows Scenario 2, where audit effectiveness and penalties are sufficiently high for an interior benchmark to exist; trajectories starting from different interior compositions move around that benchmark rather than converging monotonically to it. The full set of phase diagrams for all scenarios and values of λ is reported in the Figure Appendix.

5.3 Fiscal revenue and institutional performance

The stationary shares derived above can also be used to compute the fiscal implications of the model. This exercise does not introduce a new equilibrium concept. It shows how the same compliance–enforcement interaction is reflected in voluntary tax payments, revenue recovered through inspection, inspection costs, and the authority's institutional-performance payoff.

Let $q_C(t)=1-q_H(t)$ denote the share of cheating taxpayers and $p_{NI}(t)=1-p_I(t)$ denote the share of non-inspecting units. At each time t, voluntary tax revenue is composed of the tax paid by honest taxpayers and the tax voluntarily paid by partial evaders

$$R_{T1}(t) = \tau q_H(t) Y$$

$$R_{T2}(t) = \tau q_C(t) \beta Y.$$

Audit revenue has two components. The first is penalty revenue collected from detected evasion. The second is the evaded tax recovered through inspection

$$R_P(t) = q_C(t) p_I(t) \varphi \alpha (1-\beta) Y$$

$$R_{T3}(t) = \tau q_C(t) p_I(t) \alpha (1-\beta) Y.$$

The factor α enters both audit-revenue components, because inspection is not perfectly effective. When a cheating taxpayer is inspected, only the fraction α of evaded tax and related penalties is detected and recovered.

Voluntary tax revenue, audit revenue, and aggregate gross public revenue are therefore

$$R_{Tax}(t) = R_{T1}(t) + R_{T2}(t)$$

$$R_{audit}(t) = R_P(t) + R_{T3}(t)$$

$$R_{TOT}(t) = R_{Tax}(t) + R_{audit}(t).$$

Inspection is costly. Actual fiscal revenue net of the administrative cost of inspection is

$$R_{TOT,net}(t) = R_{TOT}(t) - p_I(t) C \tau Y.$$

This is the accounting revenue measure. It includes voluntary tax payments, recovered evaded tax, collected penalties, and the actual administrative cost of inspection. It does not treat non-inspection as a cash expenditure.

The institutional-performance payoff of the authority also accounts for recoverable but unrecovered tax losses. Define

$$F(t) = p_{NI}(t)q_C(t)\alpha\tau(1-\beta)Y.$$

F(t) measures the expected recoverable tax revenue that remains unrecovered when cheating taxpayers are not inspected. It is not an accounting cost in the public budget. It enters the authority's performance payoff only through the accountability parameter λ

$$P_A(t) = R_{TOT,net}(t) - \lambda F(t).$$

Thus, $R_{TOT,net}(t)$ measures fiscal revenue net of actual inspection costs, whereas $P_A(t)$ measures the authority's performance payoff when recoverable but unrecovered tax losses are also considered.

Table 4 reports the numerical revenue components for the four scenarios. Income is normalized to Y=1. The tax rate is τ=0.30, the declared fraction of income by cheating taxpayers is β=0.75, and the inspection-cost parameter is C=0.50. The scenarios differ in audit effectiveness α and the penalty rate φ: Scenario 1 has low α and low φ; Scenario 2 has high α and high φ; Scenario 3 has high α and low φ; Scenario 4 has low α and high φ. The full-compliance tax-revenue benchmark is τY=0.300.

Scenario 1 is not an interior case. Audit effectiveness and penalties are both too low to sustain inspection incentives. The relevant outcome is therefore a low-compliance, no-inspection boundary. Revenue comes only from the fraction of income voluntarily declared by cheating taxpayers, so R_{TOT} is 0.225, below the full-compliance benchmark of 0.300. Since no inspections are carried out, there is no audit revenue and no inspection cost. As λ increases, actual net revenue is unchanged, but

Table 4 Revenue and performance indicators by scenario

λ	Scenario	q_H	p_I	R_{Tax}	R_{audit}	R_{TOT}	Cost	F	$R_{TOT,net}$	P_A
0	Scenario 1	0.000	0.000	0.225	0.000	0.225	0.000	0.030	0.225	0.225
0	Scenario 2	0.630	0.185	0.272	0.028	0.300	0.028	0.020	0.272	0.272
0	Scenario 3	0.167	0.417	0.237	0.062	0.300	0.062	0.033	0.237	0.237
0	Scenario 4	0.167	0.417	0.237	0.062	0.300	0.062	0.015	0.237	0.237
0.5	Scenario 1	0.000	0.000	0.225	0.000	0.225	0.000	0.030	0.225	0.210
0.5	Scenario 2	0.658	0.185	0.274	0.026	0.300	0.028	0.019	0.272	0.263
0.5	Scenario 3	0.298	0.417	0.247	0.053	0.300	0.062	0.028	0.237	0.224
0.5	Scenario 4	0.231	0.417	0.242	0.058	0.300	0.062	0.013	0.237	0.231
1	Scenario 1	0.000	0.000	0.225	0.000	0.225	0.000	0.030	0.225	0.195
1	Scenario 2	0.683	0.185	0.276	0.024	0.300	0.028	0.017	0.272	0.255
1	Scenario 3	0.394	0.417	0.255	0.045	0.300	0.062	0.024	0.238	0.214
1	Scenario 4	0.286	0.417	0.246	0.054	0.300	0.062	0.012	0.237	0.225

P_A falls, because the authority's performance payoff attaches a higher institutional penalty to recoverable but unrecovered tax losses.

Scenarios 2–4 all reach the full-compliance gross revenue benchmark R_{TOT}=0.300, but they do so through different enforcement mechanisms and with different costs. Scenario 2 is the strongest governance configuration, because audit effectiveness and penalties are both high. It reaches the benchmark with a lower inspection share, p_I=0.185, and therefore with a lower inspection cost. As a result, $R_{TOT,net}$ is higher in Scenario 2 than in Scenarios 3 and 4.

Scenarios 3 and 4 show why audit effectiveness and penalties should not be treated as substitutes. In Scenario 3, α is high but φ is low; in Scenario 4, φ is high but α is low. Both cases require a higher inspection share than Scenario 2 and therefore generate lower net revenue. Their audit-revenue composition also differs. With higher audit effectiveness, the authority recovers relatively more evaded tax. With higher penalties but lower audit effectiveness, the authority relies relatively more on penalty revenue. This distinction matters, because penalty revenue is not the same as an administratively productive tax system: if α is low, less evaded tax is recovered even when the formal penalty is high.

The role of λ is also visible in the table. In the interior scenarios, increasing λ raises q_H and reduces q_C, while p_I is unchanged. Gross revenue remains equal to the full-compliance benchmark, but the composition of revenue changes: voluntary tax revenue rises and audit revenue falls. This is economically intuitive. When the authority attaches institutional value to recoverable but unrecovered tax losses, the stationary taxpayer population contains fewer cheaters. Consequently, less revenue has to be recovered through audits.

At the same time, P_A decreases when λ increases, because P_A includes the institutional-performance deduction λF(t). This should not be read as a fall in actual fiscal revenue. Actual net fiscal revenue is $R_{TOT,net}$. The lower value of P_A indicates that, under stronger accountability, residual recoverable but unrecovered tax losses receive greater weight in the authority's performance evaluation.

Values of P_A should therefore not be interpreted as welfare comparisons across different values of λ, because changing λ changes the authority's institutional objective. The comparison is mainly informative within each institutional regime and for showing how performance evaluation weights residual recoverable but unrecovered tax losses.

Figures 2, 3, 4 summarize the same revenue information graphically. The stacked bar reports R_{TOT}, decomposed into R_{Tax} and R_{audit}. The dark bar reports $R_{TOT,net}$, and the gray bar reports P_A. The figures should be read together with Table 4: the table gives the numerical values, while the figures show how the balance between voluntary payments, audit recoveries, inspection costs, and institutional performance changes across enforcement environments and values of λ.

5.4 Main numerical lessons

The numerical exercise gives an economic reading of the analytical results. The model does not imply that the system necessarily converges to a stable long-run

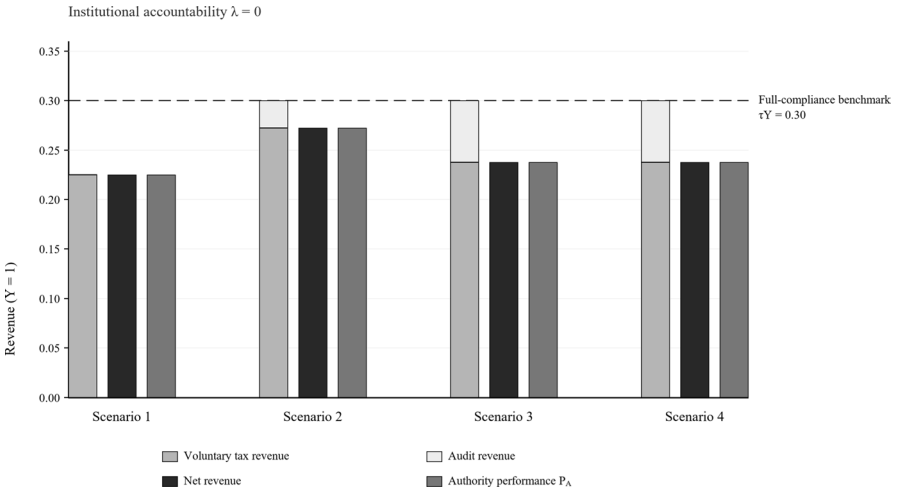


Fig. 2 Revenue components by scenario (λ=0). The stacked bar reports R_{TOT}, decomposed into R_{Tax} and R_{audit}; the dark bar reports $R_{TOT,net}$; the grey bar reports PA. The dashed horizontal line is the full-compliance tax-revenue benchmark τY=0.30

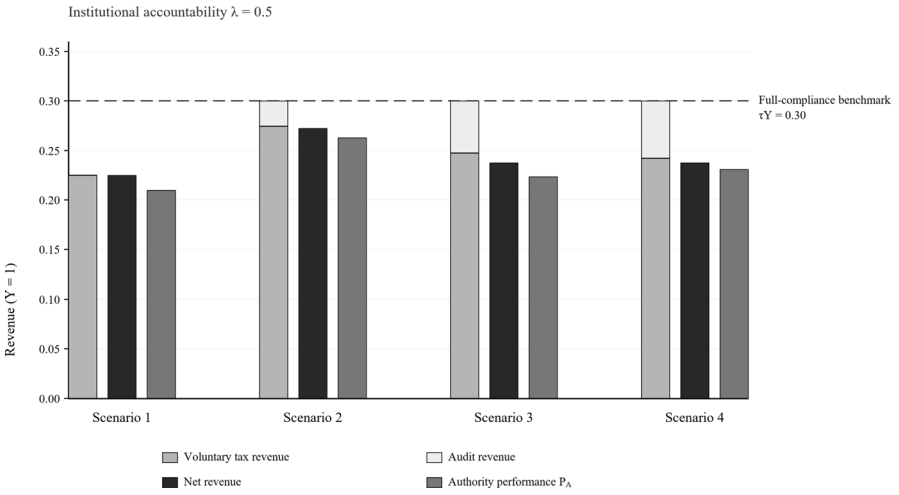


Fig. 3 Revenue components by scenario (λ=0.5). The stacked bar reports R_{TOT}, decomposed into R_{Tax} and R_{audit}; the dark bar reports $R_{TOT,net}$; the grey bar reports P_A. The dashed horizontal line is the full-compliance tax-revenue benchmark τY=0.30

point. Where the interior stationary composition exists, it identifies the central composition around which the compliance–enforcement dynamics move and, for closed interior trajectories, the corresponding time averages. Where it does not exist, the relevant outcome is a boundary regime.

The first lesson is that enforcement technology and penalty design must be considered jointly. Scenario 2, with high α and high φ, delivers the most favorable

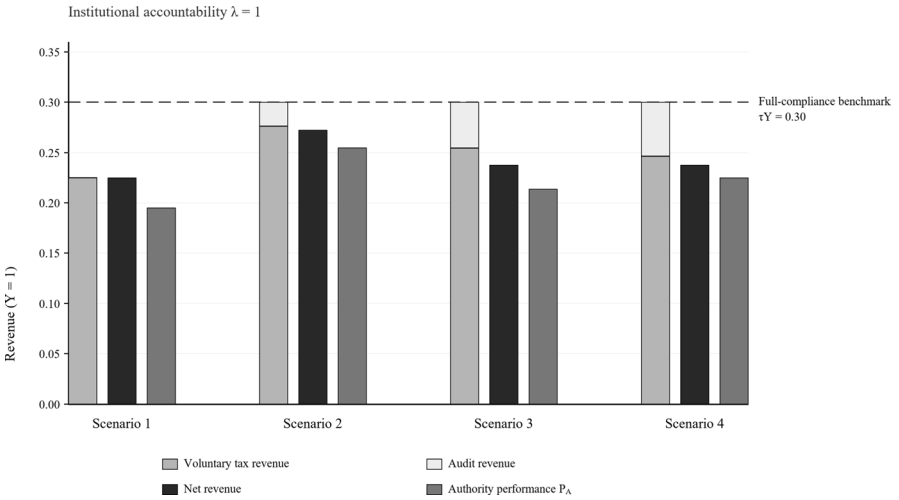


Fig. 4 Revenue components by scenario (λ=1). The stacked bar reports R_{TOT}, decomposed into R_{Tax} and R_{audit}; the dark bar reports $R_{TOT,net}$; the grey bar reports P_A. The dashed horizontal line is the full-compliance tax-revenue benchmark τY=0.30

combination: a high share of honest taxpayers, a relatively low inspection share, and the highest net fiscal revenue among the interior scenarios. This is the clearest numerical expression of the complementarity between effective recovery and meaningful penalties.

The second lesson is that a weakness on either enforcement margin is costly. Scenario 3 shows that high audit effectiveness is less powerful when penalties are weak. Scenario 4 shows that high penalties are less powerful when audit effectiveness is low. Both cases sustain an interior compliance–enforcement regime, but only with a higher inspection share and lower net revenue than Scenario 2. The policy implication is not simply to raise penalties or to increase audits, but to make inspections productive and sanctions credible at the same time.

The third lesson concerns institutional accountability. A higher λ improves the stationary compliance composition, because non-inspection becomes less attractive when recoverable but unrecovered tax losses are present. This effect operates through the authority's inspection incentive, not through the taxpayer's direct payoff. For this reason, p_I is unchanged, while q_H rises and q_C falls. In fiscal terms, a higher λ shifts revenue away from audit recoveries and toward voluntary tax payments.

The fourth lesson is that the boundary case is economically meaningful. Scenario 1 does not fail because of a numerical accident. It fails, because low audit effectiveness and low penalties make inspection insufficiently productive. The outcome is a low-compliance, no-inspection regime with lower gross and net revenue. This should be interpreted as a governance failure within the model, not as a welfare ranking of tax systems.

Overall, the numerical results support the same message as the theoretical analysis. Compliance is sustained when enforcement is productive, penalties are meaningful, and the authority has institutional reasons to act on recoverable but unrecovered

tax losses. If any of these elements is weak, the system either requires more costly inspection to sustain compliance or moves toward a boundary regime. The model therefore links fiscal performance to the governance of enforcement, rather than treating compliance as the mechanical result of statutory tax and penalty rates alone.

6 Policy implications

The model suggests that tax enforcement should be understood as a governance process, not only as a choice of statutory penalties or audit probabilities. Compliance depends on the way taxpayers and the authority adapt to each other over time. Taxpayers react to the expected payoff from honesty and under-declaration, while the authority's incentive to inspect depends on the composition of the taxpayer population and on the institutional value attached to recovering evasion.

A first implication concerns audit effectiveness. In the model, α is the effective recovery rate conditional on inspection. Policies that improve information systems, third-party reporting, data matching, audit quality, and administrative capacity can be interpreted as raising α only insofar as they increase the fraction of evaded tax and related penalties actually detected and recovered when an inspection occurs. The mechanism is therefore administrative and fiscal, not a separate belief process.

A second implication is that audit effectiveness and penalties are complements. Higher penalties can support compliance only if evasion is likely to be detected and recovered. Conversely, more effective audits have limited deterrent force when the sanction attached to detected evasion is weak. The numerical scenarios illustrate this point: the strongest compliance and net-revenue outcome is obtained when α and φ are both high.

A third implication concerns the endogenous supply of monitoring. When cheating is widespread, inspection is valuable, because the expected return to monitoring is high. When enforcement succeeds and compliance rises, the immediate return to inspection may fall. If monitoring then weakens, evasion can again become attractive. This feedback may motivate the study of visible enforcement capacity, audit routines, or other commitment devices that prevent monitoring incentives from collapsing, especially in high-compliance environments.

The accountability parameter λ captures a further institutional margin. A tax administration evaluated only on current collections is close to $\lambda=0$. An administration that also attaches performance value to recoverable but unrecovered tax losses is closer to $\lambda>0$. This does not change actual fiscal revenue mechanically; it changes the authority's payoff from non-inspection when evasion could have been detected and recovered. Performance indicators based on recoverable but unrecovered tax losses can therefore strengthen monitoring incentives without being treated as accounting costs.

Finally, the model clarifies the limits of its own policy reach. It is a random-matching model of costly monitoring, not a model of audit selection or risk-based targeting. Better targeting and digital tax-administration tools are relevant to the extent that they raise effective recovery or reduce inspection costs (OECD 2021), but the baseline model does not analyze signal-contingent audit rules. Extending the

framework to targeted audits, taxpayer belief updating, or explicit audit floors would require additional state variables and policy instruments.

7 Conclusion

This paper has analyzed tax evasion and audit activity as a co-evolutionary governance problem. Taxpayer compliance and inspection effort are not fixed policy primitives; they are population shares that adjust according to relative payoffs. This perspective highlights a feedback that is absent from static deterrence models: evasion makes inspection valuable, inspection reduces evasion, lower evasion weakens inspection incentives, and weaker inspection can reopen space for evasion.

The analysis shows that the interior benchmark of the game should not be interpreted as an automatically attracting long-run point. When the interior benchmark exists, the canonical replicator generates recurrent interior motion around it. For closed interior cycles, the benchmark gives the time averages of compliance and inspection, and the product-average identities derived above justify its use as a compact benchmark for the associated revenue components. The governance problem is therefore not simply how to choose tax rates, penalties, or audit effectiveness, so that the system reaches a fixed endpoint. It is how institutional incentives shape the supply of monitoring over the compliance cycle.

The model also distinguishes actual fiscal revenue from the authority's institutional-performance payoff. A tax administration may be evaluated not only on current cash collections but also on its ability to limit recoverable tax losses. The parameter λ captures this internalization. Higher institutional internalization of recoverable tax losses increases the stationary share of compliant taxpayers, while leaving actual fiscal revenue accounting conceptually separate from performance evaluation.

Several extensions remain. Future work could introduce risk-based targeting, explicit belief dynamics, taxpayer heterogeneity, social interaction, or data-driven audit technologies. These extensions would allow the model to speak more directly to modern tax-administration reforms. The baseline contribution is more focused: even in a simple random-matching environment, endogenous audit incentives can generate persistent compliance–enforcement cycles.

Appendix A

Comparative statics derivations

This appendix reports the derivatives underlying Table 1. All derivatives are ceteris paribus: when one parameter varies, all other parameters are held fixed. Under the interior conditions, all denominators below are positive.

A.1 Stationary inspection and non-inspection shares

$$p_I^* = \tau/[\alpha(\tau+\varphi)], p_{NI}^* = 1 - p_I^* \tag{23}$$

$$\frac{\partial p_I^*}{\partial \tau} = \varphi/\left[\alpha(\tau+\varphi)^2\right] > 0$$

$$\frac{\partial p_I^*}{\partial \alpha} = -\tau/[\alpha^2(\tau+\varphi)] < 0$$

$$\frac{\partial p_I^*}{\partial \varphi} = -\tau/[\alpha(\tau+\varphi)^2] < 0$$

$$\frac{\partial p_I^*}{\partial C} = \frac{\partial p_I^*}{\partial \beta} = \frac{\partial p_I^*}{\partial \lambda} = 0.$$

Since $p_{NI}^* = 1 - p_I^*$, the derivatives of p_{NI}^* have the opposite signs.

A.2 Stationary shares of cheating and honest taxpayers

The following derivatives refer to the stationary share of cheating taxpayers. The stationary honest share moves in the opposite direction

$$q_C^* = C\tau/[\alpha(1-\beta)(\varphi+(1+\lambda)\tau)], q_H^* = 1 - q_C^* \tag{24}$$

$$\frac{\partial q_C^*}{\partial C} = \tau/[\alpha(1-\beta)(\varphi+(1+\lambda)\tau)] > 0$$

$$\frac{\partial q_C^*}{\partial \beta} = C\tau/[\alpha(1-\beta)^2(\varphi+(1+\lambda)\tau)] > 0$$

$$\frac{\partial q_C^*}{\partial \alpha} = -C\tau/[\alpha^2(1-\beta)(\varphi+(1+\lambda)\tau)] < 0$$

$$\frac{\partial q_C^*}{\partial \varphi} = -C\tau/[\alpha(1-\beta)(\varphi+(1+\lambda)\tau)^2] < 0$$

$$\frac{\partial q_C^*}{\partial \lambda} = -C\tau^2/[\alpha(1-\beta)(\varphi+(1+\lambda)\tau)^2] < 0$$

$$\frac{\partial q_C^*}{\partial \tau} = \frac{C\varphi}{\left[\alpha(1-\beta)(\varphi+(1+\lambda)\tau)^2\right]} > 0.$$

Since $q_H^* = 1 - q_C^*$, the derivatives of q_H^* have the opposite signs.

Appendix B

Cash-revenue specification of the authority payoff

This appendix reports the full model under the alternative specification in which a non-inspecting unit matched with a cheating taxpayer receives only the cash revenue actually collected. The term representing recoverable but unrecovered tax losses is therefore removed from the authority payoff. All other elements of the model are kept unchanged: taxpayers may be honest or may under-declare income, the authority may inspect or not inspect, matching is random, inspection is costly, and inspection recovers only a fraction α of evaded tax and related penalties.

The purpose of the appendix is to show exactly what changes when the authority payoff is written as a pure fiscal-revenue payoff. The taxpayer side of the model is unchanged. The authority side changes only in the payoff from non-inspection against a cheating taxpayer. As a result, the taxpayer indifference condition and the stationary inspection share are unchanged, while the authority indifference condition is evaluated without the institutional-performance term λ.

B.1 Payoffs

Let $x=q_H$ denote the share of honest taxpayers and $1-x=q_C$ denote the share of cheating taxpayers. Let $y=p_I$ denote the share of inspecting units and $1-y=p_{NI}$ denote the share of non-inspecting units. True income is Y, the tax rate is τ, a cheating taxpayer declares the fraction β of income, audit effectiveness is α, the penalty rate is φ, and inspection costs CτY.

For an inspecting unit matched with an honest taxpayer, there is no evasion to recover. The authority receives the tax due on true income and pays the inspection cost, while the taxpayer pays the tax due

$$\pi_A^0(I,H) = \tau Y - C\tau Y, \pi_T(I,H) = (1-\tau)Y. \tag{25}$$

For an inspecting unit matched with a cheating taxpayer, the taxpayer has declared only βY. Inspection detects and recovers the fraction α of the evaded tax base and of the associated penalty base. The authority receives the tax paid on declared income, recovered tax, and the collected penalty, net of inspection cost

$$\pi_A^0(I,C) = \beta\tau Y + \alpha\tau(1-\beta)Y + \alpha\varphi(1-\beta)Y - C\tau Y. \tag{26}$$

The corresponding taxpayer payoff under inspection is

$$\pi_T(I, C) = Y - \beta\tau Y - \alpha\tau(1-\beta)Y - \alpha\varphi(1-\beta)Y. \tag{27}$$

For a non-inspecting unit matched with an honest taxpayer, the authority receives the tax paid on full income and bears no inspection cost. The taxpayer obtains the same disposable income as in the inspected honest case

$$\pi_A^0(NI, H) = \tau Y, \pi_T(NI, H) = (1-\tau)Y. \tag{28}$$

For a non-inspecting unit matched with a cheating taxpayer, the alternative cash-revenue specification removes the institutional-performance loss from the authority payoff. The authority collects only the tax voluntarily paid on declared income

$$\pi_A^0(NI, C) = \tau\beta Y. \tag{29}$$

The cheating taxpayer who is not inspected pays tax only on declared income and keeps the remaining income

$$\pi_T(NI, C) = Y - \beta\tau Y. \tag{30}$$

Equations (25)–(30) therefore define the same four matches as in the main model, but with a purely cash-based payoff for the authority under non-inspection. In particular, there is no λ term in (29).

B.2 Fitness functions and payoff differences

Expected payoffs are obtained by weighting each payoff by the composition of the other population. Under the cash-revenue specification, the authority's expected payoff from inspection is

$$f_A^0(I) = x\pi_A^0(I, H) + (1-x)\pi_A^0(I, C). \tag{31}$$

The authority's expected payoff from non-inspection is

$$f_A^0(NI) = x\pi_A^0(NI, H) + (1-x)\pi_A^0(NI, C). \tag{32}$$

The taxpayer's expected payoffs are unchanged

$$f_T(H) = y\pi_T(I, H) + (1-y)\pi_T(NI, H) \tag{33}$$

$$f_T(C) = y\pi_T(I, C) + (1-y)\pi_T(NI, C). \tag{34}$$

The taxpayer payoff difference is the same as in the main model

$$\Delta_T(y) = f_T(H) - f_T(C) = (1-\beta)Y\left[\alpha(\tau+\varphi)y - \tau\right]. \tag{35}$$

The authority payoff difference becomes

$$\Delta_A^0(x) = f_A^0(I) - f_A^0(NI) = Y\{\alpha(1-\beta)(\tau+\varphi)(1-x) - C\tau\}. \tag{36}$$

The only difference relative to the main specification is the term multiplying the cheating share in the authority condition. With the institutional-performance loss, the corresponding term is $\alpha(1-\beta)[\varphi+(1+\lambda)\tau]$. Under the cash-revenue specification, it is $\alpha(1-\beta)(\tau+\varphi)$.

B.3 Canonical replicator dynamics

The canonical two-population replicator is

$$\dot{x} = x(1-x)\Delta_T(y), \dot{y} = y(1-y)\Delta_A^0(x). \tag{37}$$

Substituting (35) and (36) gives

$$\dot{x} = x(1-x)(1-\beta)Y\left[\alpha(\tau+\varphi)y - \tau\right] \tag{38}$$

$$\dot{y} = y(1-y)Y\{\alpha(1-\beta)(\tau+\varphi)(1-x) - C\tau\}. \tag{39}$$

The system is therefore still an asymmetric two-population replicator. Boundary shares remain invariant because of the factors $x(1-x)$ and $y(1-y)$.

B.4 Interior stationary benchmark

An interior stationary point requires both payoff differences to be zero. From (35), taxpayer indifference gives

$$p_I^0 = \frac{\tau}{[\alpha(\tau+\varphi)]}. \tag{40}$$

Equivalently, using the notation of the main text

$$p_I^0 = \frac{\tau}{[\alpha(\tau+\varphi)]}, p_{NI}^0 = 1 - p_I^0. \tag{41}$$

From (36), authority indifference gives

$$q_C^0 = \frac{C\tau}{[\alpha(1-\beta)(\tau+\varphi)]}, q_H^0 = 1 - q_C^0. \tag{42}$$

The interior benchmark exists if and only if all shares lie strictly between zero and one. This requires

$$\alpha(\tau+\varphi) > \tau \tag{43}$$

and

$$\alpha(1-\beta)(\tau+\varphi) > C\tau. \tag{44}$$

Condition (43) is the taxpayer-side condition. It requires inspection to be sufficiently effective, relative to the tax saving from under-declaration, to make honesty viable. Condition (44) is the authority-side condition. It requires inspection to be productive enough, relative to its cost, to make the inspection strategy viable when cheating is present.

B.5 Comparative statics

The stationary inspection share is the same as in the main specification, because the taxpayer payoff difference is unchanged

$$p_I^0 = \frac{\tau}{[\alpha(\tau+\varphi)]} \tag{45}$$

$$\frac{\partial p_I^0}{\partial \tau} = \frac{\varphi}{\alpha(\tau+\varphi)^2} > 0, \quad \frac{\partial p_I^0}{\partial \alpha} = -\frac{\tau}{\alpha^2(\tau+\varphi)} < 0, \quad \frac{\partial p_I^0}{\partial \varphi} = -\frac{\tau}{\alpha(\tau+\varphi)^2} < 0 \tag{46}$$

$$\frac{\partial p_I^0}{\partial C} = \frac{\partial p_I^0}{\partial \beta} = 0. \tag{47}$$

The stationary cheating share under the cash-revenue specification is

$$q_C^0 = \frac{C\tau}{[\alpha(1-\beta)(\tau+\varphi)]} \tag{48}$$

$$\frac{\partial q_C^0}{\partial C} = \frac{\tau}{[\alpha(1-\beta)(\tau+\varphi)]} > 0, \frac{\partial q_C^0}{\partial \beta} = \frac{C\tau}{\left[\alpha(1-\beta)^2(\tau+\varphi)\right]} > 0 \tag{49}$$

$$\frac{\partial q_C^0}{\partial \alpha} = -\frac{C\tau}{\left[\alpha^2(1-\beta)(\tau+\varphi)\right]} < 0, \frac{\partial q_C^0}{\partial \varphi} = -\frac{C\tau}{\left[\alpha(1-\beta)(\tau+\varphi)^2\right]} < 0 \tag{50}$$

$$\frac{\partial q_C^0}{\partial \tau} = \frac{C\varphi}{\left[\alpha(1-\beta)(\tau+\varphi)^2\right]} > 0. \tag{51}$$

The stationary honest share moves in the opposite direction, because $q_H^0 = 1 - q_C^0$. Thus, higher inspection cost and a higher declared fraction by cheating taxpayers raise the cheating share needed to sustain inspection. Higher audit effectiveness and higher penalties lower it. The tax rate raises both the inspection share required to discipline taxpayers and the cheating share needed to sustain the authority's inspection incentive.

B.6 Local dynamics and first integral

Define

$$A = (1-\beta)\alpha(\tau+\varphi)Y, B_0 = \alpha(1-\beta)(\tau+\varphi)Y. \tag{52}$$

Around the interior benchmark, the system can be written as

$$\dot{x} = x(1-x)A\left(y-y^0\right),\ \dot{y} = y(1-y)B_0\left(x^0-x\right). \tag{53}$$

The Jacobian at the interior benchmark is

$$J_0\left(x^0,y^0\right) = \left[\left[0, x^0\left(1-x^0\right)A\right], \left[-y^0\left(1-y^0\right)B_0, 0\right]\right]. \tag{54}$$

Its trace is zero and its determinant is positive

$$trJ_0\left(x^0,y^0\right) = 0, \det J_0\left(x^0,y^0\right) = x^0\left(1-x^0\right)y^0\left(1-y^0\right)AB_0 > 0. \tag{55}$$

The eigenvalues are therefore purely imaginary

$$\mu_{\{1,2\}} = \pm i\sqrt{\left[x^0\left(1-x^0\right)y^0\left(1-y^0\right)AB_0\right]}. \tag{56}$$

Thus, the interior benchmark is not a locally attracting endpoint under the canonical replicator. It is the center of recurrent compliance–enforcement motion. The corresponding first integral is

$$H_0(x,y) = B_0\left[x^0\ln x + \left(1-x^0\right)\ln(1-x)\right] + A\left[y^0\ln y + \left(1-y^0\right)\ln(1-y)\right]. \tag{57}$$

Along every interior trajectory

$$\frac{dH_0}{dt} = 0. \tag{58}$$

Differentiating (57) gives

$$\frac{dH_0}{dt} = B_0\left[\frac{x^0}{x} - \frac{1-x^0}{1-x}\right]\dot{x} + A\left[\frac{y^0}{y} - \frac{1-y^0}{1-y}\right]\dot{y} \tag{59}$$

Substituting (53) gives

$$\frac{dH_0}{dt} = AB_0\left(x^0-x\right)\left(y-y^0\right) + AB_0\left(y^0-y\right)\left(x^0-x\right) = 0. \tag{60}$$

Therefore, the cash-revenue specification preserves the cyclic structure of the model. If an interior trajectory is closed, the time averages over a full cycle satisfy

$$\bar{x} = x^0,\ \bar{y} = y^0. \tag{61}$$

The benchmark is therefore still economically meaningful as the average composition of compliance and inspection over the cycle, even though the system does not converge monotonically to it.

B.7 Boundary regimes

The boundary logic is also unchanged in form. Define the two margins

$$D_0 = \alpha(\tau + \varphi) - \tau, E_0 = \alpha(1 - \beta)(\tau + \varphi) - C\tau. \tag{62}$$

When $D_0 > 0$ and $E_0 > 0$, the interior benchmark exists and the interior dynamics are recurrent. When $D_0 < 0$, inspection is not strong enough to make honesty viable in the taxpayer population, and the model moves toward a low-compliance boundary. When $E_0 < 0$, inspection is not rewarding enough to cover its cost, and the model moves toward a low-inspection boundary. Equality cases are knife-edge transitions between regimes.

Thus the formulas in (41) and (42) should be interpreted as interior benchmark values only when (43) and (44) hold. If either condition fails, the candidate values diagnose why the interior composition is not economically admissible.

B.8 Relation to the main specification

The cash-revenue specification is the case in which the authority does not internalize recoverable but unrecovered tax losses in its evolutionary payoff. The corresponding stationary shares are

$$q_C^0 = \frac{\tau C}{[\alpha(1 - \beta)(\varphi + \tau)]}, q_H^0 = 1 - q_C^0, p_I^0 = p_I^*. \tag{63}$$

In the main specification, when $\lambda>0$, the authority-side denominator becomes $\alpha(1 - \beta)[\varphi + (1 + \lambda)\tau]$, which is larger than $\alpha(1 - \beta)(\varphi + \tau)$. The corresponding stationary shares are

$$q_C^* = \frac{\tau C}{\{\alpha(1 - \beta)[\varphi + (1 + \lambda)\tau]\}}, q_H^* = 1 - q_C^*.$$

$$\text{Since } \varphi + (1 + \lambda)\tau > \varphi + \tau \text{ for } \lambda > 0, \text{ the main specification implies that}$$

$$q_C^* < q_C^0, \quad q_H^* > q_H^0 \quad \text{for } \lambda > 0. \tag{64}$$

The inspection share is unchanged, because λ does not enter the taxpayer payoff difference. The economic interpretation is straightforward. Removing the institutional-performance term makes non-inspection more attractive whenever cheating taxpayers are present. A higher cheating share is then needed to make inspection

worthwhile for the authority. The cash-revenue specification is therefore the lower-accountability benchmark of the model.

The cash-revenue specification changes the location of the interior benchmark but not the local dynamic structure of the canonical two-population replicator. Around the interior benchmark, the linearized system still has zero trace and positive determinant. Hence, the benchmark is a center rather than an attracting rest point. The absence of asymptotic stability is therefore not driven by the institutional-performance term in the authority payoff; it follows from the feedback structure of the two-population replicator.

The revenue accounting remains conceptually separate from the evolutionary payoff. Under the cash-revenue specification, actual fiscal revenue is still decomposed into voluntary tax revenue, recovered tax, penalty revenue, and inspection costs. What disappears is not a budget item, but the institutional-performance deduction from the authority's evolutionary payoff. This appendix therefore shows that the core dynamic mechanism does not depend on subtracting recoverable but unrecovered tax losses from cash revenue. That term changes the location and admissibility of the authority-side benchmark, but not the canonical replicator structure, the taxpayer condition, or the interpretation of the interior benchmark as the average composition of a cycle.

Figure appendix. Replicator phase diagrams

This appendix reports the phase diagrams used in the numerical illustration. The horizontal axis is q_H, the share of honest taxpayers, and the vertical axis is p_I, the share of inspecting units. A point in a diagram is therefore a composition of the two populations. The three colored starting points are the same in every figure: a balanced composition, (q_H, p_I)=(0.50,0.50); high compliance with low inspection, (0.80,0.20); and low compliance with high inspection, (0.20,0.80).

For Scenarios 2–4, the colored lines are the closed trajectories passing through those starting points. They represent recurrent compliance–enforcement movements around the stationary benchmark, marked by a star. These curves are drawn as complete closed trajectories, rather than as truncated time simulations, so the reader can see the full cycle selected by each initial condition. Scenario 1 is different: because audit effectiveness and the penalty are both low, no admissible stationary benchmark exists. The corresponding figures therefore show movement toward a low-compliance, low-inspection boundary.

The scenario labels report the calibration parameters. The parameter α is audit effectiveness: it measures the fraction of evaded tax and related penalties that an inspection can detect and recover. The parameter φ is the penalty rate. The parameter λ is institutional accountability: it measures how much the authority internalizes recoverable but unrecovered tax losses in its performance payoff. The figures are grouped by λ, while the four scenarios vary α and φ (Figs. 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16).

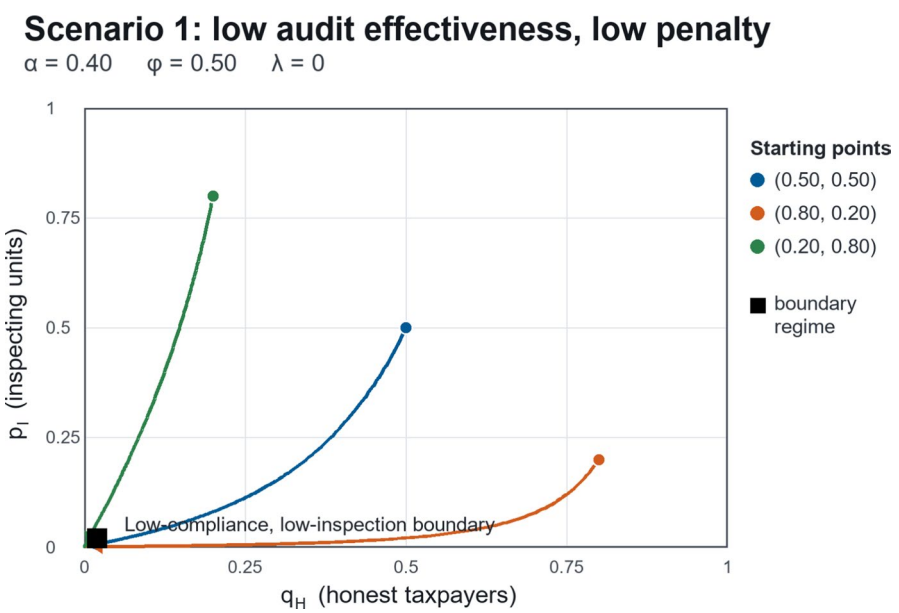


Fig. 5 Scenario 1, λ=0, α=0.40, φ=0.50: no admissible stationary benchmark; trajectories move toward the low-compliance, low-inspection boundary. The black square marks the boundary direction

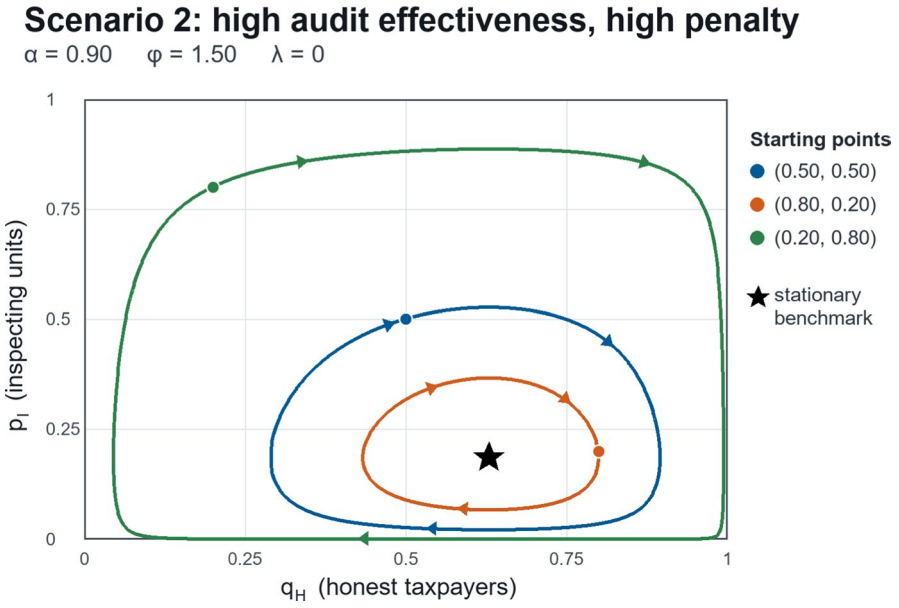


Fig. 6 Scenario 2, λ=0, α=0.90, φ=1.50: strong enforcement; closed trajectories around the stationary benchmark. The star marks the stationary benchmark. $q_H^* = 0.630, p_I^* = 0.185$

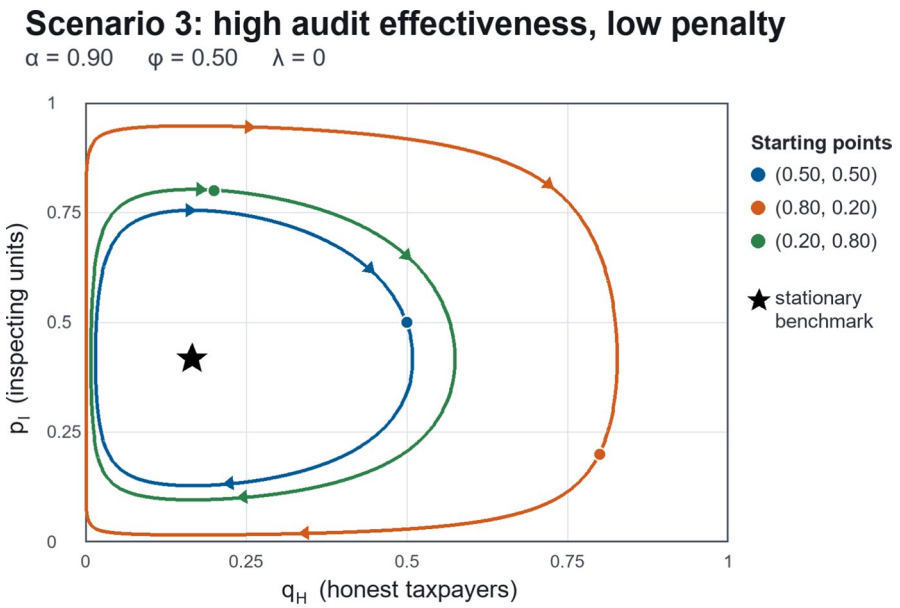


Fig. 7 Scenario 3, λ=0, α=0.90, φ=0.50: effective audits but low penalty; closed trajectories around a weaker compliance benchmark. The star marks the stationary benchmark. $q_H^* = 0.167, p_I^* = 0.417$

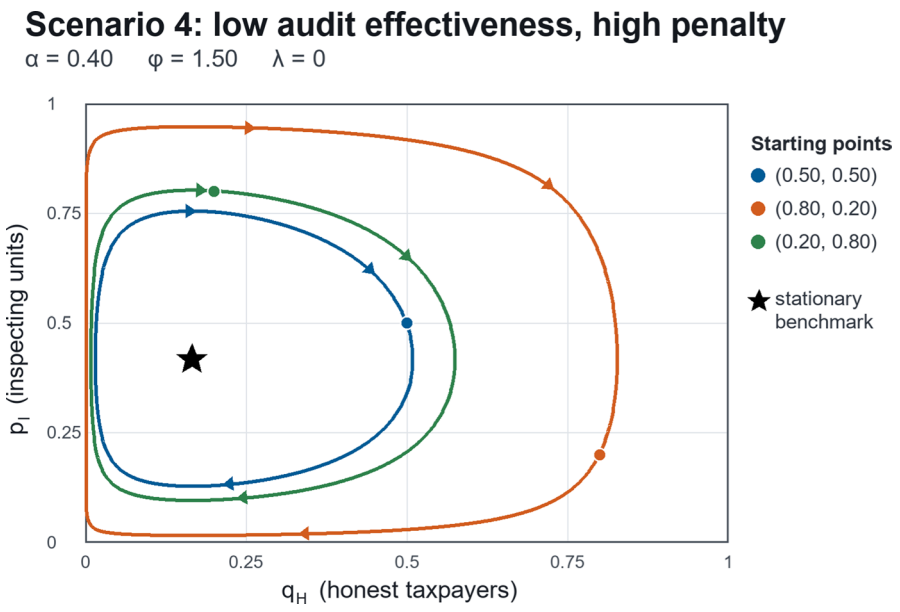


Fig. 8 Scenario 4, λ=0, α=0.40, φ=1.50: high penalty but lower audit effectiveness; closed trajectories with lower compliance than Scenario 2. The star marks the stationary benchmark. $q_H^* = 0.167, p_I^* = 0.417$

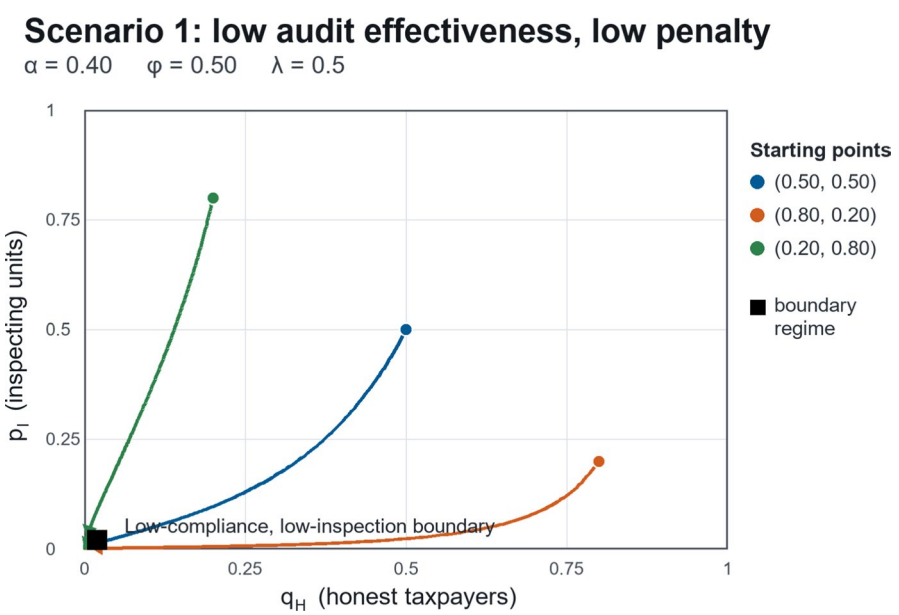


Fig. 9 Scenario 1, λ=0.5, α=0.40, φ=0.50: no admissible stationary benchmark; trajectories move toward the low-compliance, low-inspection boundary. The black square marks the boundary direction

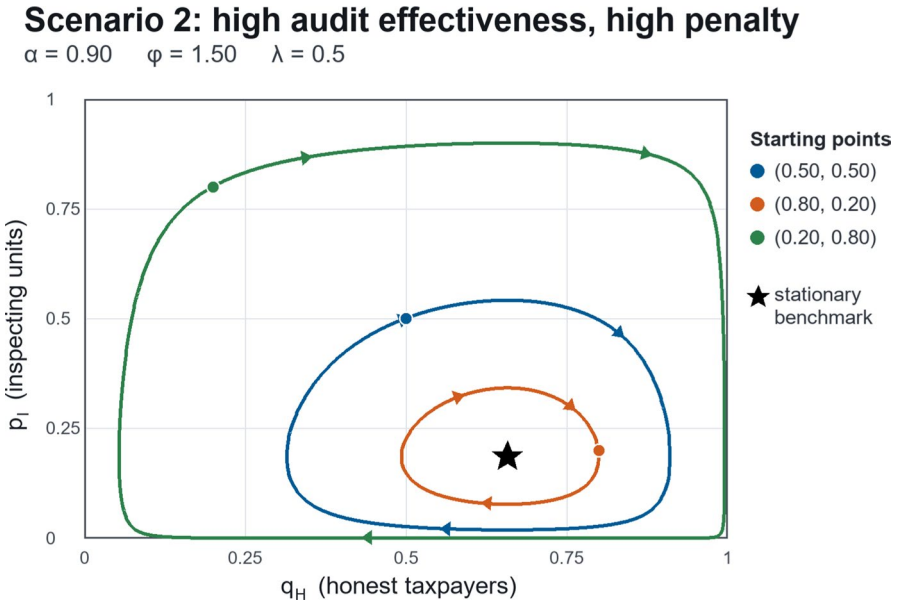


Fig. 10 Scenario 2, λ=0.5, α=0.90, φ=1.50: strong enforcement; closed trajectories around the stationary benchmark. The star marks the stationary benchmark. $q_H^* = 0.658, p_I^* = 0.185$

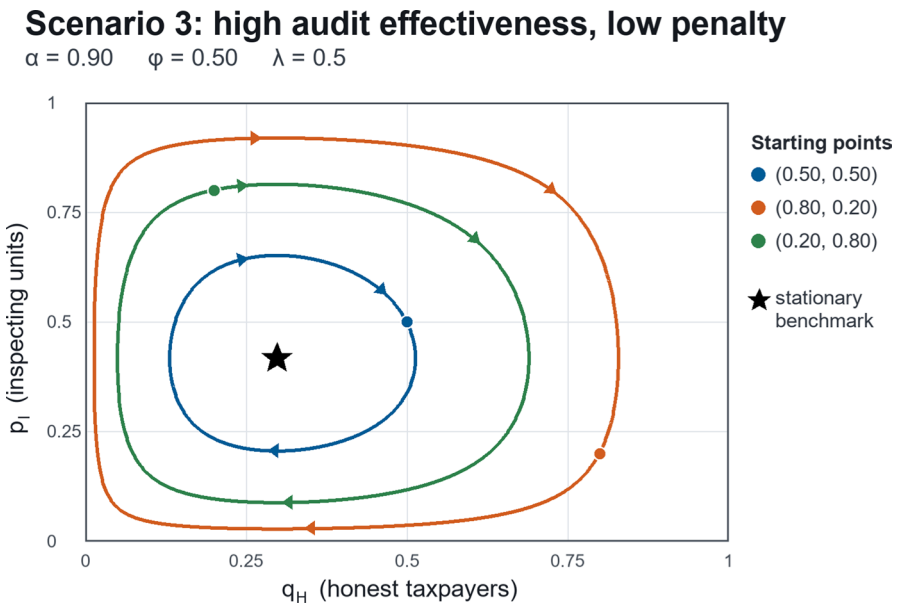


Fig. 11 Scenario 3, λ=0.5, α=0.90, φ=0.50: effective audits but low penalty; closed trajectories around a weaker compliance benchmark. The star marks the stationary benchmark. $q_H^* = 0.298, p_I^* = 0.417$

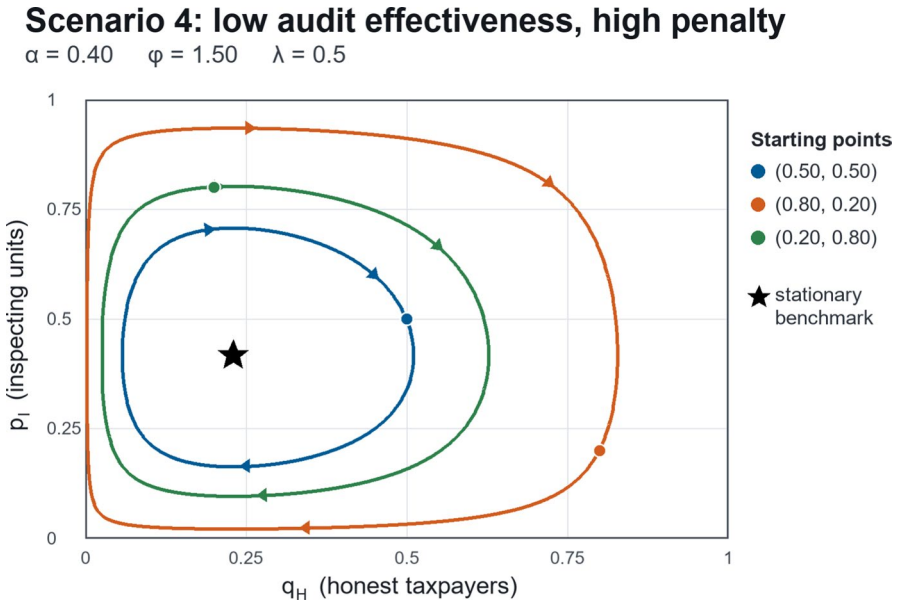


Fig. 12 Scenario 4, λ=0.5, α=0.40, φ=1.50: high penalty but lower audit effectiveness; closed trajectories with lower compliance than Scenario 2. The star marks the stationary benchmark. $q_H^* = 0.231, p_I^* = 0.417$

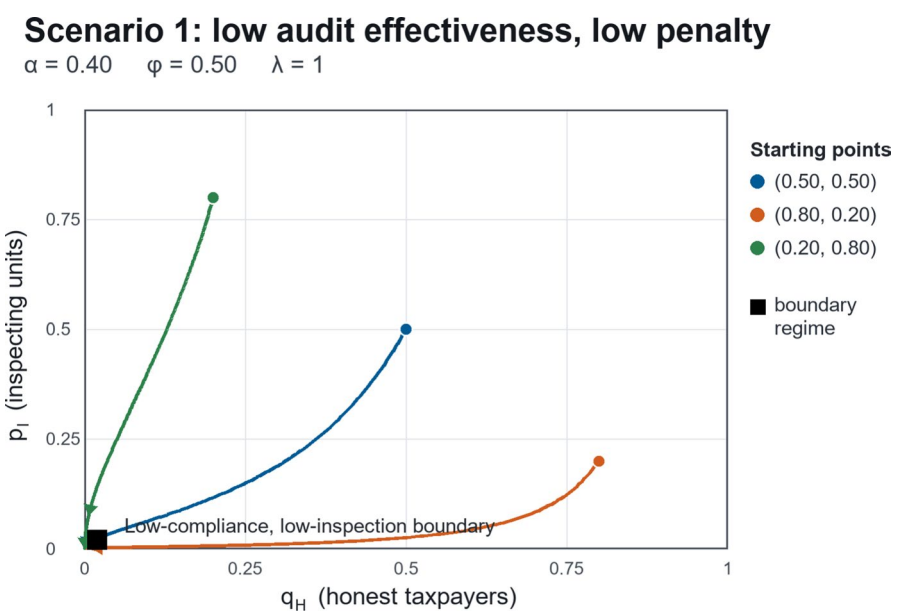


Fig. 13 Scenario 1, λ=1, α=0.40, φ=0.50: no admissible stationary benchmark; trajectories move toward the low-compliance, low-inspection boundary. The black square marks the boundary direction

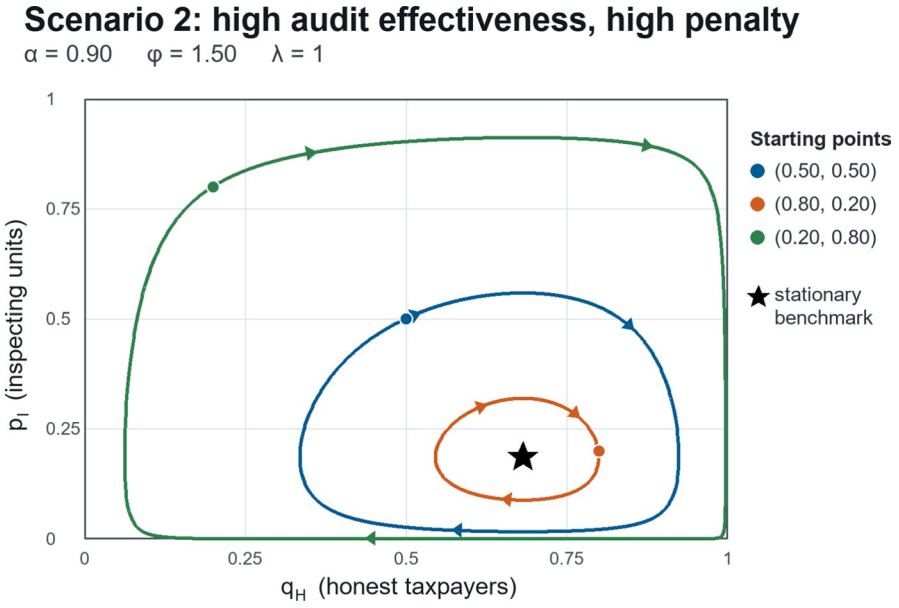


Fig. 14 Scenario 2, λ=1, α=0.90, φ=1.50: strong enforcement; closed trajectories around the stationary benchmark. The star marks the stationary benchmark. $q_H^* = 0.683, p_I^* = 0.185$

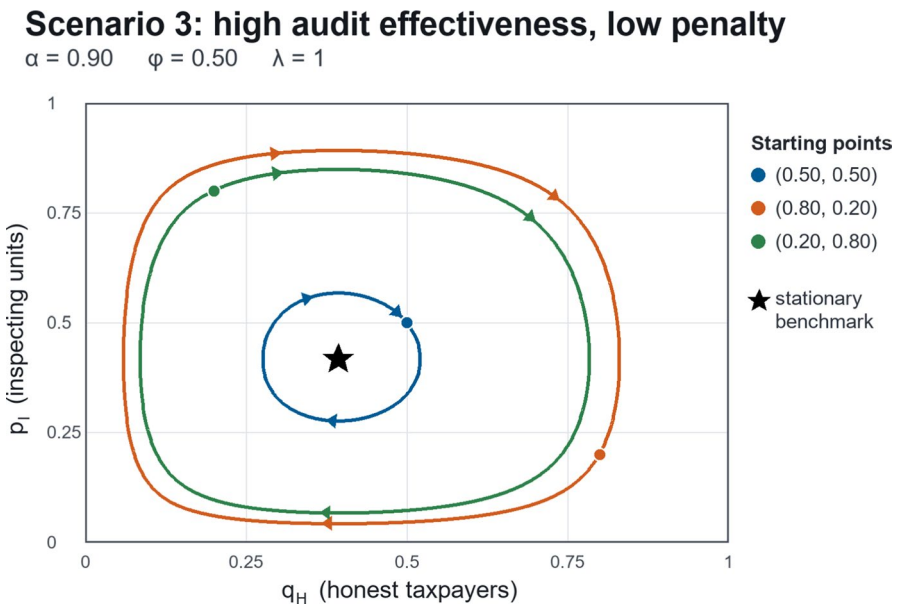


Fig. 15 Scenario 3, λ=1, α=0.90, φ=0.50: effective audits but low penalty; closed trajectories around a weaker compliance benchmark. The star marks the stationary benchmark. $q_H^* = 0.394, p_I^* = 0.417$

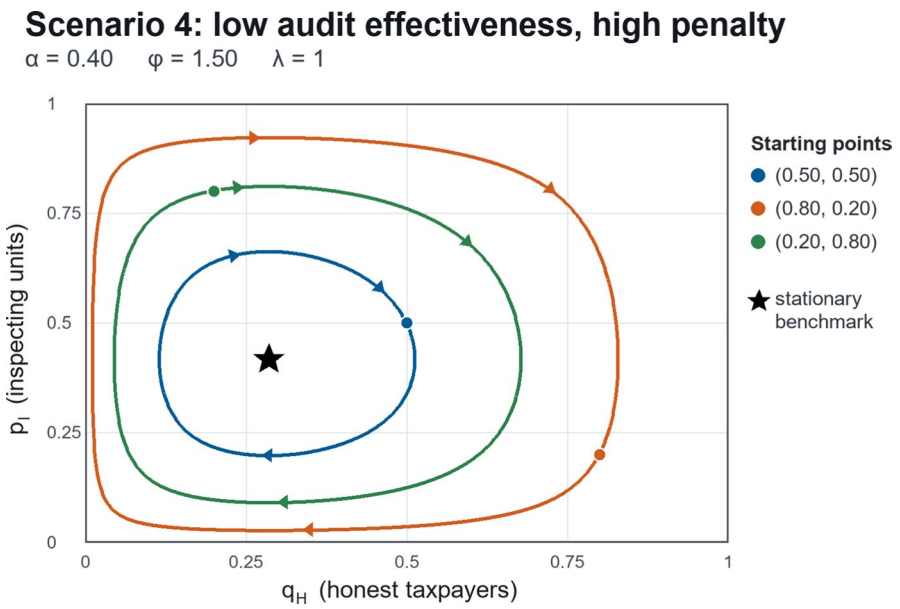


Fig. 16 Scenario 4, λ=1, α=0.40, φ=1.50: high penalty but lower audit effectiveness; closed trajectories with lower compliance than Scenario 2. The star marks the stationary benchmark. $q_H^* = 0.286, p_I^* = 0.417$

Author contributions S.F. wrote the final manuscript text. E.C. and S.F. addressed Section 2–4. S.F. addressed Section 5–7 and the appendix.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest None.

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